# UNIT - I

## Energy Theorems and Deflections

## 1. Introduction

- Structural analysis finds **forces (axial, shear, bending moment, torsion)** and **deformations (deflection, slope)** in a structure under loads.
- There are two broad approaches: **Force (flexibility) methods** and **Displacement (stiffness) methods** . Energy methods give a powerful tool for both.
- **Energy methods** use the principle: *work done by external loads = strain energy stored in the structure* (for a conservative, linear elastic system).
- Where equilibrium equations alone give deflection only with difficulty (trusses, frames, curved beams), energy theorems give a quick and systematic answer.

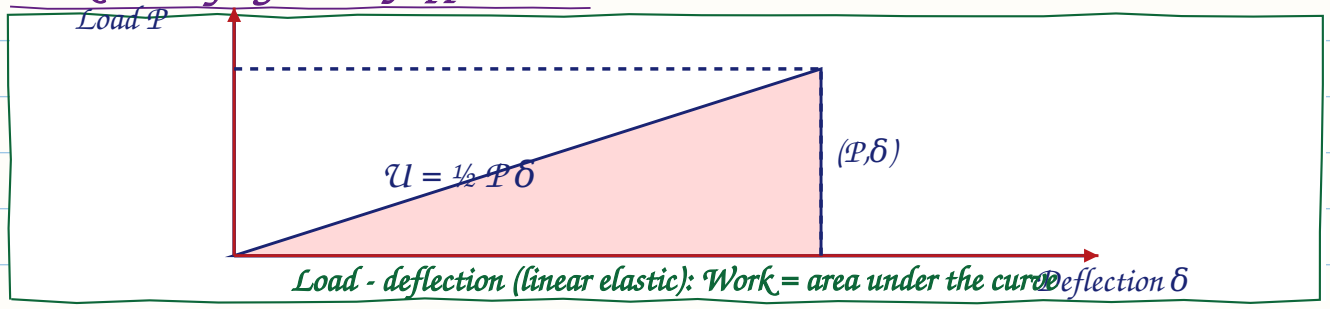
**Basic assumptions**

1. Material is **linearly elastic** (Hooke's law valid) 2. Deflections are **small** 3. Supports are unyielding (unless stated) 4. Principle of **superposition** holds 5. Loads are applied **gradually** (no dynamic effect).

## 2. Work, Energy and Strain Energy

- **Work** = force × displacement in the direction of the force.
- **Strain energy (U)** is the energy stored in a deformed elastic body due to the work done by external loads. It is recovered completely when the load is removed (elastic).

### Work done by a gradually applied load




- Work done = area of the triangle = **½ P δ** . This is the strain energy stored.
- For a suddenly applied load, work = P δ — twice as much, which is why the sudden-load stress is double.

**Important terms**

**Resilience** = total strain energy stored in the body. **Proof resilience** = max strain energy stored at the elastic limit. **Modulus of resilience** = proof resilience per unit volume = $\sigma^2 / 2E$.

## 3. Strain Energy in a Linear Elastic System

- **Clapeyron's theorem:** For a linear elastic structure under several loads $P_1, P_2, \ldots$ producing deflections $\delta_1, \delta_2, \ldots$ in the direction of the loads (all applied gradually together):

**Clapeyron's theorem of work**

$U = \frac{1}{2}(P_1\delta_1 + P_2\delta_2 + \ldots) = \frac{1}{2}\Sigma P\delta$

- Strain energy is always **positive** and is a **quadratic function** of load ($U \propto P^2$). So **superposition does not apply to strain energy** — you cannot add $U$ of two load cases separately.
- Strain energy per unit volume under direct stress: $u = \sigma^2/2E = \frac{1}{2}\sigma\varepsilon$ .

## 4. Expressions for Strain Energy

### (a) Due to axial load

- For a bar of length $L$, area $A$, modulus $E$ carrying axial force $P$:

**Axial load**

$U = P^2 L/2AE$ (uniform bar)

$U = \int P^2\,dx/2AE$ (when $P$ or $A$ varies along length)

$U = \Sigma P_i^2 L_i/2A_iE_i$ (bar of several segments)

- Derivation: Stress $\sigma = P/A$, strain $\varepsilon = P/AE$; $u = \sigma^2/2E$; $U = u \times$ volume $= (P^2/2A^2E) \times AL = P^2L/2AE$.

### (b) Due to bending moment

- Consider an element $dx$ of a beam with bending moment $M$. Rotation between the two faces $d\theta = M\,dx/EI$. Work $= \frac{1}{2}M\,d\theta$.

**Bending**

$U = \int M^2\,dx/2EI$ (integrate over the full length of the member)

- Derivation: Since: $dU = \frac{1}{2}M\,d\theta = \frac{1}{2}M \times (M\,dx/EI) = M^2\,dx/2EI$.

### (c) Due to shear force

- Average shear stress $\tau = V/A$; strain energy per unit volume $= \tau^2/2G$. A form factor $\alpha$ accounts for non-uniform shear stress across the section.

**Shear**

$U = \int \alpha V^2\,dx/2GA$ where $\alpha$ = form factor for shear

- $\alpha = 6/5 = 1.2$ for rectangle, $10/9$ for solid circle, about **1.0 for I-section** (area of web only), **2.0** for thin tube.

### (d) Due to torsion (for completeness)

**Torsion**

$U = \int T^2\,dx / 2GJ$ (for a shaft: $U = T^2\,L / 2GJ$)

### Total strain energy in a member

**Total**

$U = \int N^2\,dx / 2AE + \int M^2\,dx / 2EI + \int \alpha V^2\,dx / 2GA + \int T^2\,dx / 2GJ$

→ In beams, the shear and axial energies are very small compared to bending energy, so only $U = \int M^2\,dx / 2EI$ is used. In trusses, only axial energy exists. In frames, axial + bending (shear neglected).

| Type of member | Energy to consider | Expression |
|---|---|---|
| Beam | bending (shear neglected) | $\int M^2\,dx / 2EI$ |
| Pin-jointed truss | axial only | $\Sigma F^2\,L / 2AE$ |
| Rigid-jointed frame | bending (+ axial if needed) | $\int M^2\,dx / 2EI$ |
| Curved bar / ring | bending + axial + shear | all three terms |
| Shaft | torsion | $T^2\,L / 2GJ$ |

## 5. Strain Energy due to Impact and Sudden Load (extra points)

- **Gradual load P:** stress $\sigma = P/A$; $U = \tfrac{1}{2} P \delta$.
- **Sudden load:** stress = $2P/A$ (double); extension = $2P\,L / AE$.
- **Impact load** (weight $W$ falling through height $h$ on a bar of length $L$):

**Instantaneous stress**

$\sigma = (W/A)[\,1 + \sqrt{(\,1 + 2\,h\,A\,E / W\,L\,)}\,]$ ; if $h = 0$ then $\sigma = 2W/A$.

## 6. Castigliano's Theorems

- Proposed by **Carlo Alberto Castigliano** (Italian engineer, 1873).

**Castigliano's first theorem (deflection form, as in most Indian syllabi)**

*"The partial derivative of the total strain energy of a structure, with respect to any one of the applied loads, gives the displacement of the point of application of that load, in the direction of that load."*

$\delta = \partial U / \partial P$ and for a moment: $\theta = \partial U / \partial M$

- Here $U$ must be written in terms of the loads (as a function of $P$). $P$ can be a force (gives linear deflection) or a couple (gives rotation).

**Note on naming**

Some books call this the **Castigliano's second theorem** ($\delta = \partial U/\partial P$) and call the "force = $\partial U/\partial \delta$" form the first theorem. In our syllabus: **Castigliano's I = deflection theorem ($\delta = \partial U/\partial P$)** and **Castigliano's II = theorem of least work ($\partial U/\partial R = 0$)**, used in Unit II.

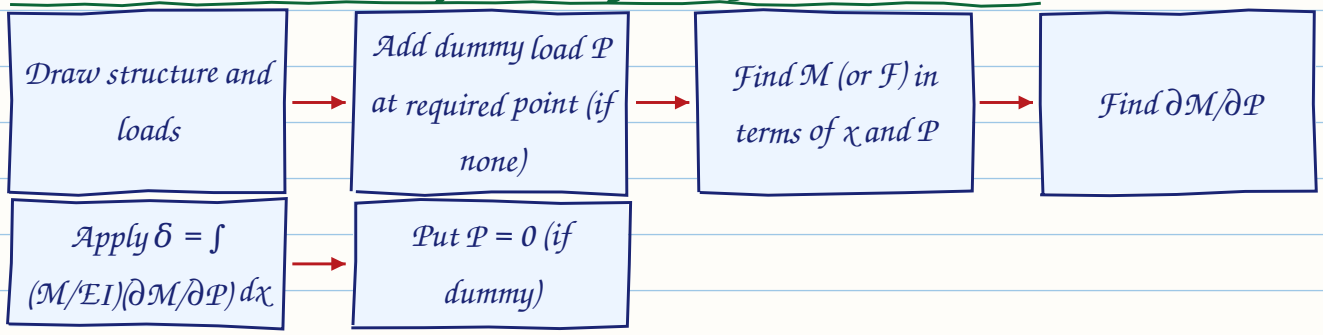
### Proof (short)

- Loads $P_1 \dots P_n$ act and the strain energy is $U$. Let $P_n$ increase by $dP_n$: the strain energy increases by $dU = (\partial U/\partial P_n)\, dP_n$.
- Now apply $dP_n$ **first** and then the original loads: the energy is $\tfrac{1}{2}\, dP_n\, d\delta_n + \delta_n\, dP_n + U$. Energy stored does not depend on the order of loading, so equate both and neglect the second-order term $\tfrac{1}{2}\, dP_n\, d\delta_n$.
- Hence $\partial U/\partial P_n = \delta_n$.

### Conditions for use

- Elastic, linear system; supports do not move; temperature constant.
- The deflection is to be found **at the point and in the direction of an existing load**. If there is no load at that point → **apply a dummy (fictitious) load $P$** there, find $\partial U/\partial P$, then put $P = 0$.

## 7. Procedure to Find Deflection by Castigliano's Theorem




1. Write the bending moment $M$ in each segment (with limits of $x$) including $P$.
2. **Differentiate first, integrate later** — it avoids squaring: $\partial U/\partial P = \partial/\partial P \int M^2 dx/2EI = \int (M/EI)(\partial M/\partial P)\, dx$.
3. Integrate segment by segment and add.
4. Substitute $P = 0$ for the dummy load. A **positive** answer means deflection is in the direction of $P$.

- For a rotation: apply a dummy moment $M_0$ at the point; $\theta = \int (M/EI)(\partial M/\partial M_0)\, dx$ with $M_0 = 0$.

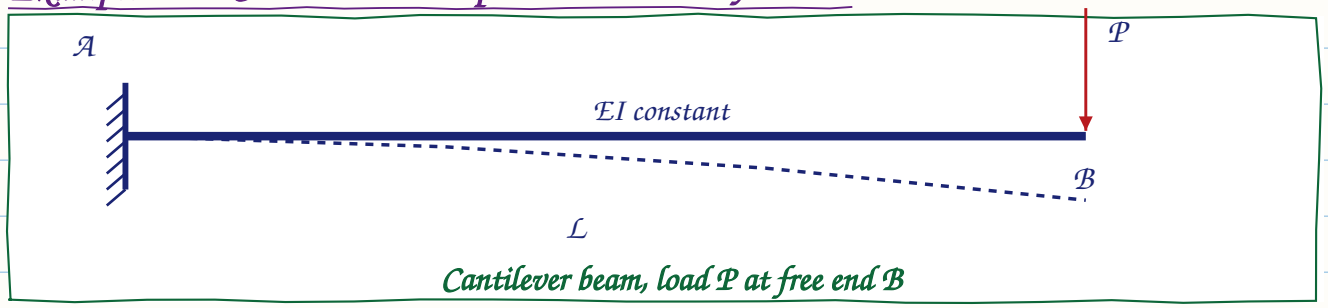
**Formula to remember**

**Beams:** $\delta = \int (M/EI)(\partial M/\partial P)\, dx$

**Trusses:** $\delta = \Sigma (FL/AE)(\partial F/\partial P)$

# 8. Deflection of Simple Beams — Worked Examples

## Example 1 — Cantilever with point load at the free end



Cantilever beam, load P at free end B

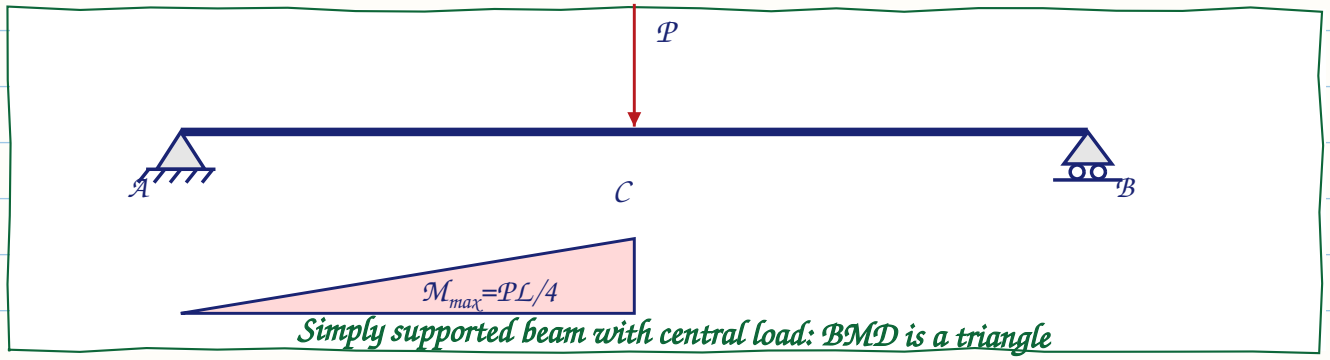
- Take $x$ from the free end. $M = -P\,x$ (hogging); $\partial M/\partial P = -x$.
- $U = \int_0^L P^2x^2\,dx / 2EI = P^2L^3/6EI$ .
- $\delta_B = \partial U/\partial P = P\,L^3/3EI$ (downwards).
- Using dummy moment $M_0$ at $B$: $M = -Px - M_0$, $\partial M/\partial M_0 = -1$; $\theta_B = \int (Px)(1)/EI\,dx = P\,L^2/2EI$ .

## Example 2 — Cantilever with UDL w over full span (deflection at free end)

- Free end has no point load, so add a **dummy load P** at $B$.
- $M = -(P\,x + w\,x^2/2)$; $\partial M/\partial P = -x$.
- $\delta_B = (1/EI)\int_0^L (P\,x + w\,x^2/2)\,x\,dx$, put $P = 0 \rightarrow (1/EI)\int w\,x^3/2\,dx = w\,L^4/8EI$ .

## Example 3 — Simply supported beam, central point load P

- Reactions $P/2$ each. For $0 \le x \le L/2$: $M = P\,x/2$; $\partial M/\partial P = x/2$.
- Symmetric, so double the first half: $\delta_C = (2/EI)\int_0^{L/2} (Px/2)(x/2)\,dx = (P/2EI) \times (L/2)^3/3 = P\,L^3/48EI$ .



Simply supported beam with central load: BMD is a triangle

- $M_{max} = P\,L/4$ at $C$.

## Example 4 — Simply supported beam with UDL (central deflection)

- Add a dummy load $P$ at midspan $C$. Reaction $R_A = wL/2 + P/2$.
- $0 \le x \le L/2$: $M = (wL/2 + P/2)\,x - w\,x^2/2$; $\partial M/\partial P = x/2$.
- $\delta_C = (2/EI)\int_0^{L/2} (\,wLx/2 - w\,x^2/2\,)(\,x/2\,)\,dx$ $(P = 0)$
- $= (1/EI)\,[\,wL\,x^3/6 - w\,x^4/8\,]_0^{L/2} = (w/EI)(\,L^4/48 - L^4/128\,)$

**Result**

$\delta_C = 5\,w\,L^4 / 384\,EI$ (downward)

### Example 5 — Numerical

**Problem**

A cantilever of span 3 m carries a point load 20 kN at the free end. $EI = 5 \times 10^4$ kN·m². Find the deflection and slope at the free end.

- $\delta = P\,L^3 / 3EI = 20 \times 27 / (3 \times 50000) = 540 / 150000 = 0.0036$ m = **3.6 mm** .
- $\theta = P\,L^2 / 2EI = 20 \times 9 / (2 \times 50000) = 0.0018$ rad = **0.103°** .

## 9. Standard Results (Remember These)

| Beam and loading | Max deflection $\delta$ | Max slope $\theta$ |
|---|---|---|
| Cantilever, point load $P$ at free end | $P\,L^3 / 3EI$ | $P\,L^2 / 2EI$ |
| Cantilever, UDL $w$ over full span | $w\,L^4 / 8EI$ | $w\,L^3 / 6EI$ |
| Cantilever, couple $M_0$ at free end | $M_0\,L^2 / 2EI$ | $M_0\,L / EI$ |
| S.S. beam, central point load $P$ | $P\,L^3 / 48EI$ | $P\,L^2 / 16EI$ |
| S.S. beam, UDL $w$ over full span | $5\,w\,L^4 / 384EI$ | $w\,L^3 / 24EI$ |
| S.S. beam, couple $M_0$ at one end | $M_0\,L^2 / 9\sqrt{3}\,EI$ | $\theta_A = M_0\,L / 3EI$, $\theta_B = M_0\,L / 6EI$ |
| S.S. beam, point load $P$ at distance $a$ from A ($b = L - a$) | under the load: $P\,a^2\,b^2 / 3EI\,L$ | $\theta_A = P\,a\,b\,(L + b) / 6EI\,L$ |

## 10. Deflection of Pin-Jointed Trusses

- In a pin-jointed truss each member carries only **axial force $F$** . So strain energy:

**Truss strain energy**

$U = \Sigma\, F^2\,L / 2AE$ (sum over all members)

- Castigliano: $\delta = \partial U / \partial P = \Sigma\, F\,(\partial F / \partial P)\,L / AE$ .
- $\partial F / \partial P = k$ = force in the member due to a **unit load** at the joint and in the direction of the required deflection (where $P$ is the load applied there, real or dummy).

**Working formula for trusses**

$\delta = \Sigma\, F\,k\,L / AE$ ($F$ = force due to actual loads, $k$ = force due to unit load)

- Tension +ve, compression –ve (the product $F\,k$ is positive when both have the same sign).
- If the load acts at the joint, use that load $P$ itself as the variable: $F = F_0 + kP$ form.

### Procedure

Find member forces $F$ (method of joints) → Remove loads, apply unit load at required joint and direction → Find $k$ in each member → Tabulate $F$, $k$, $L$, $A$, $E$

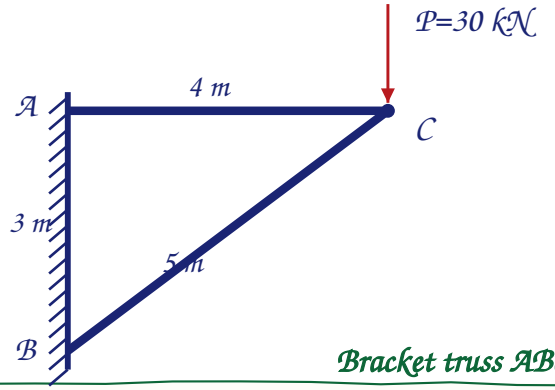
$\delta = \Sigma F k L / AE$

- For a **horizontal deflection** apply a horizontal unit (dummy) load; for vertical, a vertical unit load.
- Zero-force members ($F = 0$ and $k = 0$) contribute nothing, but may carry $k$ even if $F = 0$.

### Example 6 — Bracket truss

**Problem**

Two-member truss: $AC$ horizontal (4 m), $BC$ inclined (5 m), $A$ and $B$ on a wall 3 m apart. A vertical load $P = 30$ kN acts at $C$. $AE = 1 \times 10^5$ kN for both members. Find the vertical deflection of $C$.




Bracket truss ABC with load P at C

- Joint $C$: $F_{BC} \sin\theta = P$ with $\sin\theta = 3/5 \rightarrow F_{BC} = 50$ kN (compression); $F_{AC} = F_{BC} \cos\theta = 40$ kN (tension).
- Unit vertical load at $C$: $k_{BC} = -5/3$, $k_{AC} = +4/3$.

| Member | $F$ (kN) | $k$ | $L$ (m) | $FkL$ |
|---|---|---|---|---|
| $AC$ | +40 | +4/3 | 4 | 213.33 |
| $BC$ | −50 | −5/3 | 5 | 416.67 |
| Total | | | | 630 |

- $\delta_C = \Sigma F k L / AE = 630 / (1 \times 10^5) = 0.0063$ m = **6.3 mm** (downward).

## 11. Dummy Load Method for Trusses — Points

- Used for: vertical deflection of the bottom chord joints, horizontal movement of a roller joint, relative displacement between two joints (apply equal and opposite unit loads along the line joining them).

- **Lack of fit / temperature:** extra term $\Sigma\, k(\lambda)$ where $\lambda$ = member elongation due to temperature ($\alpha\, T\, L$) or lack of fit; $\delta = \Sigma\, F\, k\, L/AE + \Sigma\, k\lambda$ .
- Check: if loads are applied at several joints, $F = F_0 + P\, k$ form; $\partial F/\partial P$ with $P$ the particular load.

→ A common exam mistake: forgetting the sign of $F$ and $k$. Always write tension as + and compression as −.

## 11A. More Worked Examples on Beams

### Example 7 — S.S. beam with an eccentric point load




S.S. beam, point load $P$ at distance $a$ from $A$ ($b = L - a$)

- Reactions: $R_A = P\, b / L$, $R_B = P\, a / L$.
- Segment $AC$ ($0 \le x \le a$ from $A$): $M = P\, b\, x / L$. Segment $BC$ ($0 \le x' \le b$ from $B$): $M = P\, a\, x' / L$.
- $U = (1/2EI)\,[\int_0^a (P\, b\, x/L)^2\, dx + \int_0^b (P\, a\, x'/L)^2\, dx'] = P^2 a^2 b^2 (a + b) / 6EI\, L^2 =$ **$P^2 a^2 b^2 / 6EI\, L$** .
- Deflection under the load: $\delta_C = \partial U/\partial P =$ **$P\, a^2 b^2 / 3EI\, L$** .
- Check: for $a = b = L/2 \rightarrow P\, L^3 / 48EI$ √ (Example 3).

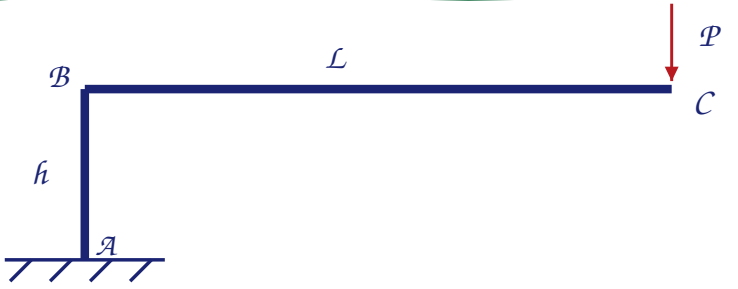
### Example 8 — Strain energy of an S.S. beam with UDL

- $M = wLx/2 - wx^2/2 \rightarrow \int_0^L M^2\, dx = w^2 L^5 / 120$.
- **$U = w^2 L^5 / 240\, EI$** .

### Example 9 — Cantilever carrying UDL and a point load at the free end

- $M = -(P\, x + w\, x^2/2)$. Using the superposition of the **deflections** (not of energies):
- **$\delta_B = P\, L^3 / 3EI + w\, L^4 / 8EI$** .

### Example 10 — L-shaped bracket (frame) — vertical deflection of C




Bracket ABC fixed at A, load P at C

- Member CB (x from C): $M = -P\,x$; $\partial M/\partial P = -x$. Member BA (y from B): $M = -P\,L$ (constant); $\partial M/\partial P = -L$.
- $\delta_C = (1/EI)\,[\int_0^L P\,x^2\,dx + \int_0^h P\,L^2\,dy\,] = P\,L^3/3EI + P\,L^2\,h\,/\,EI$.

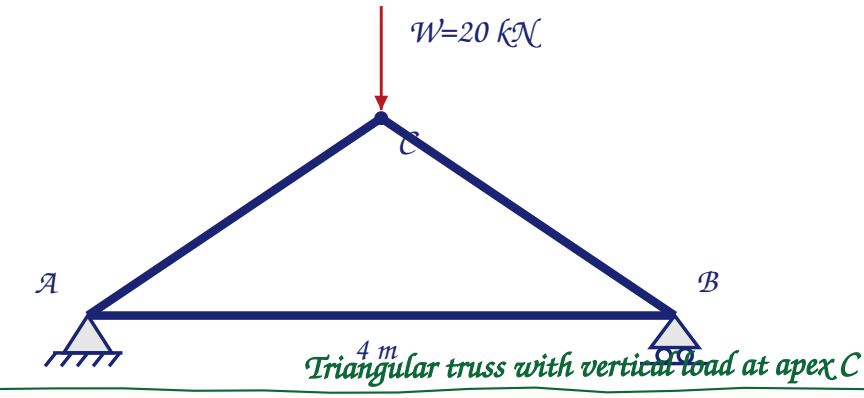
**Result**

**$\delta_C = P\,L^2\,(L + 3h)\,/\,3EI$** (axial energy of AB neglected)

- Same EI is assumed for both members. If axial effect in AB is included add $P\,h\,/\,AE$.

## 11B. More Worked Examples on Trusses

### Example 11 — Triangular truss (span 4 m, rise 3 m, load 20 kN at apex)




Triangular truss with vertical load at apex C

- $R_A = R_B = 10$ kN. Length $AC = BC = \sqrt{(2^2 + 3^2)} = \sqrt{13} = 3.606$ m.
- Joint A: $F_{AC}\sin\varphi = 10$ with $\sin\varphi = 3/\sqrt{13} \rightarrow$ **$F_{AC} = F_{BC} = -12.02$ kN** (compression); **$F_{AB} = +6.67$ kN** (tension).
- Unit vertical load at C: $k_{AC} = k_{BC} = -0.601$, $k_{AB} = +0.333$.

| Member | F (kN) | k | L (m) | F k L |
|---|---|---|---|---|
| AC | −12.02 | −0.601 | 3.606 | 26.05 |
| BC | −12.02 | −0.601 | 3.606 | 26.05 |
| AB | +6.67 | +0.333 | 4.00 | 8.89 |
| Total | | | | 60.99 |

- With $A = 1000$ mm² and $E = 200$ GPa: $AE = 2 \times 10^5$ kN $\rightarrow$ **$\delta_C = 60.99 / 2\times10^5 = 3.05 \times 10^{-4}$ m ≈ 0.305 mm** (downward).
- Horizontal movement of roller B: apply a horizontal unit load at B: only AB gets $k = +1$; $\delta_B =$

$(6.67)(1)(4) / 2\times10^5 = 0.133$ mm.

## 11C. Strain Energy — Numerical and Extra Concepts

### Example 12 — Stepped bar

**Problem**

A steel bar 2 m long: first 1 m has $A = 400$ mm², next 1 m has $A = 200$ mm². $P = 40$ kN axial; $E = 200$ GPa. Find strain energy and elongation.

- $U = (P^2 / 2E)( L_1/A_1 + L_2/A_2 ) = [ (40\times10^3)^2 / (2 \times 200\times10^9) ] \times (1/400\times10^{-6} + 1/200\times10^{-6})$
- $= 0.004 \times 7500 =$ **30 N·m (J)**.
- Elongation: since $U = \frac{1}{2} P \delta \rightarrow \delta = 2U/P = 60/40000 =$ **1.5 mm**.

### Shear energy is small — comparison for a cantilever (rectangular section b × d)

- $U_{bending} = P^2 L^3 / 6EI$; $U_{shear} = \alpha P^2 L / 2GA = 3 P^2 L / 5GA$ ($\alpha = 6/5$).
- Ratio $U_{shear} / U_{bending} = (3/10)(E/G)(d/L)^2 \approx 0.78 (d/L)^2$ (taking $E/G = 2.6$).
- For $d/L = 1/10$, shear energy is only **0.78%** of bending energy → neglected for slender beams. It matters for deep beams ($d/L \geq 1/4$).

### Castigliano's theorem in its force form

- If $U$ is expressed in terms of **displacements** $\Delta$: $P = \partial U / \partial \Delta$ (force corresponding to a displacement). Example: bar $U = \frac{1}{2} (AE/L)\Delta^2 \rightarrow P = AE\Delta / L$ √. This is the basis of **stiffness (displacement) methods** like slope-deflection (Unit IV).

### Advantages and limitations of energy methods

| Aspect | Advantages | Limitations |
|---|---|---|
| Scope | works for beams, trusses, frames, arches | only for linear elastic behaviour |
| Calculation | no need of the deflected shape | integration may be long for complex M |
| Deflection at any point | by using a dummy load | extra analysis with the dummy load |
| Indeterminate analysis | leads to least-work method (Unit II) | supports must be unyielding |

## 11D. Comparison of Methods of Finding Deflection

| Method | Basis | Best suited for |
|---|---|---|
| Double integration | $EI\, d^2y/dx^2 = M$ | simple beams, single loading |
| Macaulay's method | integration with singularity terms | beams with many loads |
| Moment-area method | area of M/EI diagram | slope and deflection at a few points |

| Method | Basis | Best suited for |
|---|---|---|
| Conjugate beam method | M/EI as load on a conjugate beam | prismatic or varying EI beams |
| Castigliano / unit load | strain energy, partial derivative | trusses, frames, any point |
| Virtual work | external virtual work = internal | general structures |

## 11E. Common Mistakes and Exam Tips

- Differentiate **before** integrating — saves effort and errors.
- A dummy load P must be **set to zero** after differentiation.
- Use correct **limits** for each segment; the origin of x may differ per segment.
- **Sign** of M need not be consistent across segments, but M and $\partial M/\partial P$ must follow the same sign convention in the same segment.
- State the **direction** of the deflection (downward/upward) in the final answer.
- In trusses write a **table** (Member, L, F, k, F k L) and total it — easy marks.
- Write the **units** : EI in kN·m², AE in kN, results in mm.

→ Quick formula: cantilever $\delta = PL^3/3EI$, UDL cantilever = $wL^4/8EI$, S.S. centre P = $PL^3/48EI$, S.S. UDL = $5wL^4/384EI$ — these may be used to check your Castigliano results.

## 12. Maxwell's Reciprocal Theorem and Betti's Law (supporting)

- **Betti's law:** In a linear elastic body, work done by a first system of forces on the displacements caused by a second system = work done by the second system on the displacements caused by the first. $P_1\delta_{12} = P_2\delta_{21}$ ($\delta_{12}$ = deflection at 1 due to load at 2).
- **Maxwell's reciprocal theorem:** The deflection at point A due to a unit load at B equals the deflection at B due to a unit load at A. $\delta_{AB} = \delta_{BA}$ (also for rotations and mixed).
- Application: influence lines by Müller–Breslau, and symmetry of flexibility and stiffness matrices.

## 13. Principle of Virtual Work / Unit Load Method (link)

- Unit load method gives the same result as Castigliano: $\delta = \int M\,m\,dx/EI$ where m = BM due to a unit load at the point and in the direction of $\delta$.
- **Castigliano** $(\partial M/\partial P) = m$ when the load is varied: so both methods coincide. Castigliano's advantage: for loads that already act at the point, no new analysis needed.

## 14. Quick Revision — Unit I

### Standard deflections

$U_{axial} = P^2 L / 2AE$ | $U_{bending} = \int M^2 dx / 2EI$ | $U_{shear} = \int \alpha V^2 dx / 2GA$ | $U_{torsion} = T^2 L / 2GJ$ | Castigliano: $\delta = \partial U/\partial P$, $\theta = \partial U/\partial M$ | Beam: $\delta = \int (M/EI)(\partial M/\partial P)\, dx$ | Truss: $\delta = \Sigma F k L/AE$.

Cantilever: $PL^3/3EI$, $wL^4/8EI$ | S.S.: $PL^3/48EI$, $5wL^4/384EI$.

### Possible exam questions

1. Derive the expressions for strain energy due to axial load, bending moment and shear force.
2. State and prove Castigliano's first theorem.
3. Find the deflection at the free end of a cantilever with a UDL using Castigliano's theorem.
4. Determine the central deflection of a simply supported beam carrying a UDL.
5. Find the vertical deflection of a joint of the pin-jointed truss given (numerical).
6. State Maxwell's reciprocal theorem.

# UNIT - II

## Analysis of Indeterminate Structures

## 1. Introduction

- A structure is **statically determinate** if all reactions and internal forces can be found using only the **equations of static equilibrium** .
- A structure is **statically indeterminate (redundant)** if the equilibrium equations are **not sufficient** ; extra equations from **compatibility of deformation** (geometry) are needed.
- **Equilibrium equations (plane structures):** $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M = 0$ — three equations.

| Basis | Determinate | Indeterminate |
|---|---|---|
| Analysis | equilibrium equations only | equilibrium + compatibility (material property EI, AE needed) |
| Forces depend on | loads and geometry only | also on stiffness (EI, AE) of members |
| Temperature / settlement / lack of fit | no stresses developed | stresses are developed |
| Failure | local failure → collapse | redistribution of moments; more safety |
| Material / sections | larger sections needed | economical, smaller sections |
| Deflection | larger | smaller (more stiff) |
| Examples | S.S. beam, cantilever, 3-hinged arch | fixed beam, continuous beam, propped cantilever, fixed portal frame |

### Advantages of indeterminate structures

- Smaller bending moments, so **economical sections** ; **stiffer** with small deflections.
- **Redundancy gives safety** — load redistributes if one member fails.
- Better stability and **more aesthetic** (slender) designs; used in bridges and multistorey buildings.

### Disadvantages

- Complex analysis; sensitive to **support settlement, temperature, lack of fit** (they induce stresses).
- Skilled design and construction needed; connections are costly.

## 2. Static Indeterminacy ($D_s$)

- **Degree of static indeterminacy = number of unknown forces – number of equilibrium equations** = number of **redundants** (extra reactions or members that can be removed without making the structure unstable).
- **External indeterminacy ($D_{se}$):** reactions exceed the available equilibrium equations.
- **Internal indeterminacy ($D_{si}$):** member forces cannot be found even if all reactions are known.
- **Total $D_s = D_{se} + D_{si}$.**

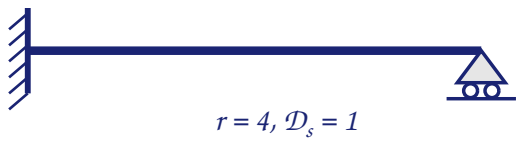
### Formulae

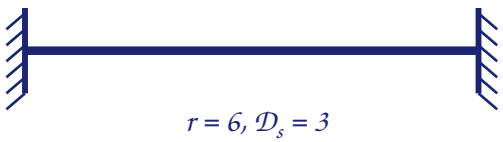
| Structure | Degree of static indeterminacy |
|---|---|
| Plane truss (pin-jointed) | $D_s = m + r - 2j$ ($m$ = members, $r$ = reaction components, $j$ = joints) |
| Plane truss, external | $D_{se} = r - 3$ |
| Plane truss, internal | $D_{si} = m - (2j - 3)$ |
| Plane rigid frame | $D_s = 3m + r - 3j$ (– number of released conditions c) |
| Beam (no hinges) | $D_s = r - 3$ |
| Beam with internal hinges | $D_s = r - 3 - e$ ($e$ = extra condition equations, e.g. hinge: 1) |
| Space truss | $D_s = m + r - 3j$ |
| Space frame | $D_s = 6m + r - 6j$ |

- For **closed loops** (rings): each closed box adds **3** degrees of indeterminacy.
- For **beams under vertical loads only**, the horizontal reaction is zero so one equation less is needed: a fixed beam is then taken as **2** degrees.

→ Stability check: reactions must not be all parallel or concurrent; $D_s$ negative means the structure is unstable (mechanism).

### Examples of static indeterminacy




**Propped cantilever ($D_s = 1$) and Fixed beam ($D_s = 3$)**

| Structure | Calculation | $D_s$ |
|---|---|---|
| Simply supported beam | $r = 3$; $3 - 3$ | 0 |
| Cantilever | $r = 3$; $3 - 3$ | 0 |
| Propped cantilever | fixed (3) + roller (1) = 4; $4 - 3$ | 1 |

| Structure | Calculation | $D_s$ |
|---|---|---|
| Fixed beam | 3 + 3 = 6; 6 − 3 | 3 (2 for vertical loads) |
| Two-span continuous beam (pin + 2 rollers) | 2 + 1 + 1 = 4; 4 − 3 | 1 |
| Three-span continuous beam (pin + 3 rollers) | 5 − 3 | 2 |
| Portal frame, hinged at both bases | 3m + r − 3j = 9 + 4 − 12 | 1 |
| Portal frame, fixed at both bases | 9 + 6 − 12 | 3 |
| Three-hinged arch/frame | r = 4, e = 1: 4 − 3 − 1 | 0 |
| Plane truss with 6 members, 4 joints, r = 3 | 6 + 3 − 8 | 1 (internal) |

- For continuous beam with n supports (one pin, rest rollers): $D_s = n - 2$ .

## 3. Kinematic Indeterminacy ($D_k$) — Degrees of Freedom

- **Degree of kinematic indeterminacy** = number of **independent joint displacements (translations and rotations)** that must be known to describe the deformed shape. It is the number of unknowns in the **displacement (stiffness) method** .
- A **fixed support** has no displacement; a **pin** can rotate; a **free end** can translate and rotate.

### Formulae

| Structure | $D_k$ |
|---|---|
| Plane truss | $D_k = 2j - r$ (each free joint has 2 translations) |
| Plane frame (axial deformation considered) | $D_k = 3j - r$ |
| Plane frame, axially rigid members | $D_k = 3j - r - m$ |
| Continuous beam (axial effect neglected) | $D_k = 2j - r$ or number of joint rotations + translations |
| Space truss | $D_k = 3j - r$ |
| Space frame | $D_k = 6j - r$ |

| Structure | Free displacements | $D_k$ | $D_s$ |
|---|---|---|---|
| Fixed beam | none | 0 | 3 |
| Propped cantilever | rotation at the pinned/roller end | 1 | 1 |

| Structure | Free displacements | $D_k$ | $D_s$ |
|---|---|---|---|
| Cantilever | rotation + deflection at free end (axial neglected) | 2 | 0 |
| Two-span continuous beam, simple supports | rotation at A, B, C | 3 | 1 |
| Two-span continuous beam, A fixed | rotation at B, C | 2 | 2 |
| Portal frame fixed bases (axial neglected) | rotation at B, C + sway | 3 | 3 |
| Portal frame hinged bases (axial neglected) | rotation at A, B, C, D + sway | 5 | 1 |
| Portal frame fixed bases (axial included) | u, v, θ at B and C | 6 | 3 |

→ Rule: Moment distribution/slope-deflection need $D_k$ unknowns; force methods (Castigliano, flexibility) need $D_s$ unknowns. Choose the method with fewer unknowns.

### Static vs Kinematic indeterminacy

| Basis | Static | Kinematic |
|---|---|---|
| Unknowns | redundant forces | joint displacements |
| Method used | force / flexibility | displacement / stiffness |
| Examples of methods | Castigliano II, consistent deformation, three-moment | slope-deflection, moment distribution, stiffness matrix |
| Fixed beam | 3 (or 2) | 0 |
| Cantilever | 0 | 2 |
| Equations | compatibility | equilibrium of joints |

## 4. Methods of Analysis of Indeterminate Structures

**Force (flexibility) methods**
Castigliano II, consistent deformation, theorem of three moments, column analogy, flexibility matrix

**Displacement (stiffness) methods**
slope-deflection, moment distribution, Kani's method, stiffness matrix

**Approximate methods**
portal method, cantilever method (for lateral loads)

**Energy and others**
unit load method, virtual work, finite element

- **Common idea:** choose the redundants → make the structure determinate (**released / primary**

**structure** ) → find displacement of the primary structure at the redundant locations (due to loads and due to unit redundants) → impose compatibility conditions.

## 5. Castigliano's Second Theorem (Theorem of Least Work)

**Statement**

**"For a statically indeterminate structure, the partial derivative of the total strain energy with respect to each redundant force (reaction or internal force) is zero, provided the supports are unyielding and no temperature or lack-of-fit effects are present."**

$\partial U / \partial R = 0$ ($R$ = redundant)

- It means: **of all the stress distributions that satisfy equilibrium, the actual one is that which makes the strain energy minimum** — hence "least work".

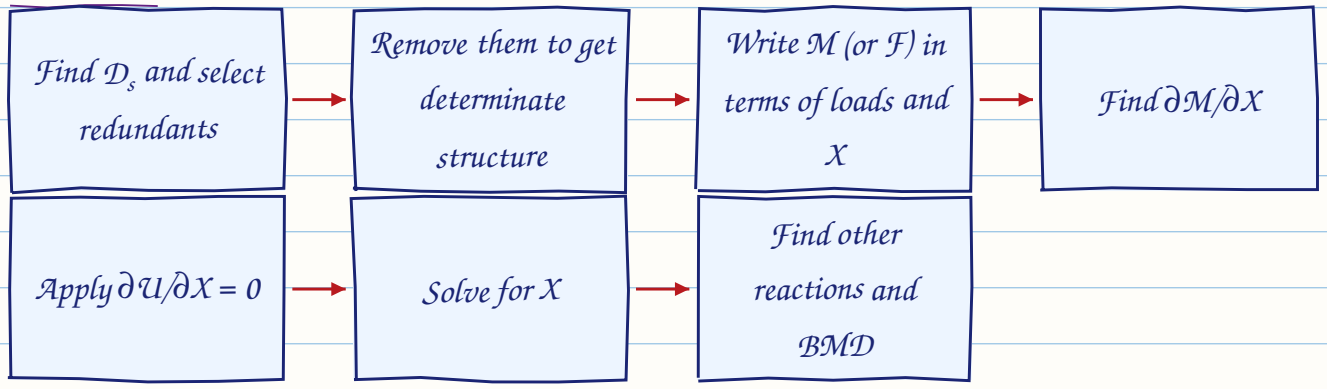
### Proof (simple)

- From Castigliano's first theorem the deflection at the point of the redundant reaction $R$ is $\delta = \partial U/\partial R$.
- Because the support is unyielding, the deflection in the direction of $R$ is **zero** ; hence $\partial U/\partial R = 0$ .
- If the support settles by $\Delta$ in the direction of $R$, then $\partial U/\partial R = \Delta$ (equal to the settlement). Temperature change and lack of fit add extra terms (see section 9).

### Equations

- For a beam: $\partial U/\partial R = \int (M/EI)(\partial M/\partial R)\,dx = 0$ .
- For a truss: $\partial U/\partial X = \Sigma (F L/AE)(\partial F/\partial X) = 0$ with $F = F_0 + kX$.
- With **n** redundants $X_1 \ldots X_n$: write n equations $\partial U/\partial X_i = 0$ → solve simultaneously.

### Procedure

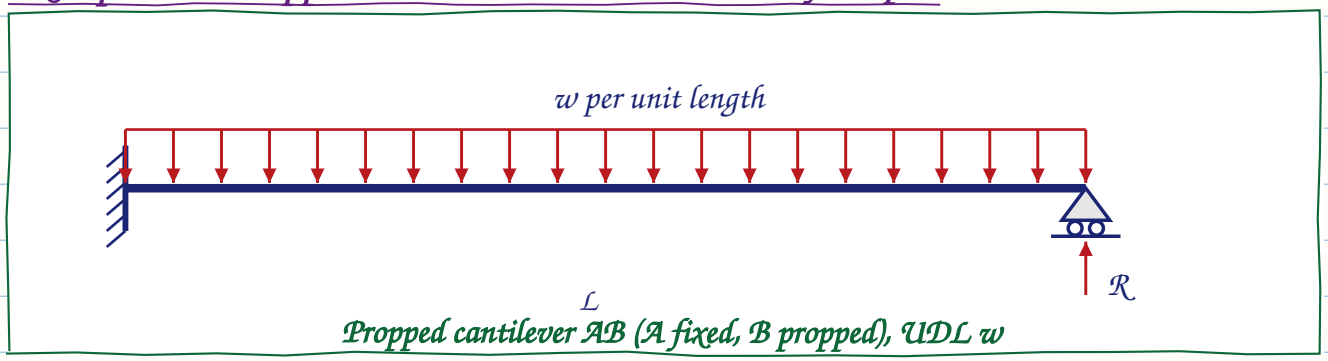



1. Determine the degree of indeterminacy and select redundant(s) — pick those that make the work easy (e.g., a prop reaction).
2. Express reactions, $M$ or $F$ in terms of the **external loads and redundants** .
3. Form $U$ and differentiate: $\partial U/\partial X = 0$ (differentiate first, integrate later).
4. Solve; then find the remaining reactions by equilibrium, then SFD and BMD.

→ Not valid if supports sink or if there is temperature change / lack of fit unless the extra terms are included.

## 6. Indeterminate Beams by Castigliano II — Examples

### Example 1 — Propped cantilever with UDL w over full span




Propped cantilever AB (A fixed, B propped), UDL w

- Take $R_B = R$ as redundant. Measure $x$ from B. $M = Rx - w x^2/2$; $\partial M/\partial R = x$.
- $\partial U/\partial R = (1/EI)\int_0^L (Rx - w x^2/2)\, x\, dx = 0 \rightarrow R L^3/3 - w L^4/8 = 0$.

**Result**

$R_B = 3wL/8$, $R_A = 5wL/8$, $M_A = -wL^2/8$ (hogging), max sagging $M = 9wL^2/128$ at $x = 3L/8$ from B.

### Example 2 — Propped cantilever with a central point load P

- $x$ from B: for $0 \le x \le L/2$, $M = Rx$; for $L/2 \le x \le L$, $M = Rx - P(x - L/2)$; $\partial M/\partial R = x$.
- $\int M x\, dx = R L^3/3 - P\int_{L/2}^{L} (x - L/2)\, x\, dx = R L^3/3 - 5PL^3/48 = 0$.

**Result**

$R_B = 5P/16$, $R_A = 11P/16$, $M_A = -3PL/16$, $M_C$ (under load) $= +5PL/32$.

### Example 3 — Two-span continuous beam, equal spans L, UDL w on both spans

- Support B (middle) is taken as redundant: $R_B = R$; by symmetry $R_A = R_C = wL - R/2$.
- Span AB ($x$ from A): $M = (wL - R/2)x - w x^2/2$; $\partial M/\partial R = -x/2$.
- $\partial U/\partial R = 2\int_0^L M(-x/2)\, dx/EI = 0 \rightarrow (wL - R/2)L^3/3 = wL^4/8$.

**Result**

$R_B = 5wL/4$, $R_A = R_C = 3wL/8$, $M_B = -wL^2/8$, max sagging $= 9wL^2/128$ at $3L/8$ from the end supports.

- Check: $R_A + R_B + R_C = 3wL/8 + 5wL/4 + 3wL/8 = 2wL$ ✓ (total load).

### Example 4 — Numerical

**Problem**

A propped cantilever 6 m long carries a UDL of 20 kN/m over the entire span. Find the prop reaction and fixed-end moment.

- $R_B = 3wL/8 = 3 \times 20 \times 6/8 = 45$ kN; $R_A = 120 - 45 = 75$ kN.

- $M_A = -wL^2/8 = -20 \times 36 / 8 = \mathbf{-90\ kN{\cdot}m}$ .

### Effect of settlement of the prop (support B sinks by Δ)

- Compatibility: deflection of the cantilever at $B = wL^4/8EI - RL^3/3EI = \Delta$ (settlement).
- Hence $\mathbf{R_B = 3wL/8 - 3EI\Delta/L^3}$ — settlement of the prop **reduces** the prop reaction.

## 7. Indeterminate Pin-Jointed Trusses

### Types of redundancy in trusses

- **External redundancy:** extra support reactions ($r > 3$). Redundant = a reaction. Eg: hinged supports at both ends ($r = 4$).
- **Internal redundancy:** more members than needed ($m > 2j - 3$). Redundant = force in a **member** . Eg: a panel with both diagonals.
- **Degree = $(m + r) - 2j$** . In this unit, up to **two** redundants (internal and/or external).

### Method (Castigliano II for trusses)

- Select the redundant(s) $X_1$ (and $X_2$). For internal redundants **cut the member** and apply $X$ on both sides of the cut. For external redundants remove the support.
- Member forces: $\mathbf{F = F_0 + k_1 X_1 + k_2 X_2}$
- $F_0$ = force in the **primary (determinate) truss** due to actual loads. $k_1$, $k_2$ = force due to a **unit** value of $X_1$, $X_2$ (the cut member itself has $k = 1$).

**Two redundants (simultaneous equations)**

$$\partial U/\partial X = \Sigma\,(F_0 + kX)\,kL/AE = 0 \Rightarrow X = -\Sigma\,F_0\,kL/AE \div \Sigma\,k^2 L/AE$$

$$X_1 \Sigma\,k_1^2 L/AE + X_2 \Sigma\,k_1 k_2 L/AE = -\Sigma\,F_0 k_1 L/AE$$

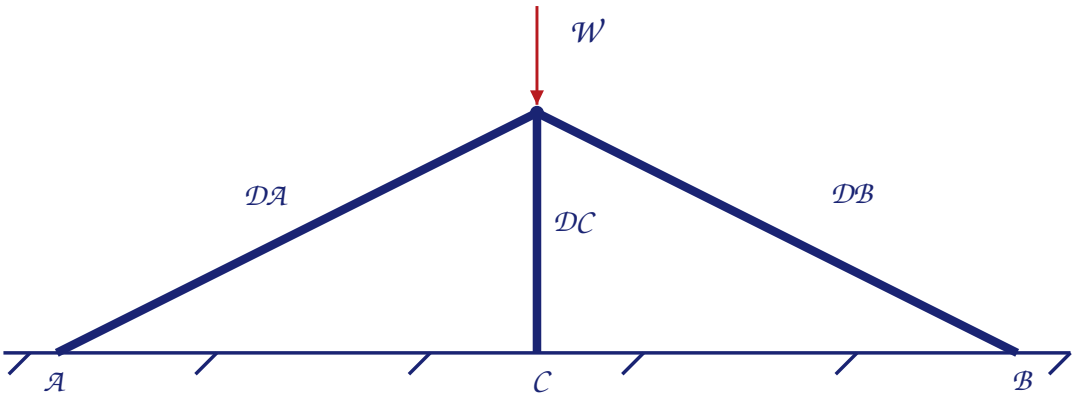
$$X_1 \Sigma\,k_1 k_2 L/AE + X_2 \Sigma\,k_2^2 L/AE = -\Sigma\,F_0 k_2 L/AE$$

- If $AE$ is same for all members it cancels out. The redundant member itself is also included in the summations.
- **Lack of fit λ** in a redundant member and **temperature change** appear as: $\Sigma\,(F_0 + kX)\,kL/AE = -\Sigma\,k\lambda$ ($\lambda = \alpha T L$ or the lack of fit; sign as per convention).

Select redundants → Find $F_0$ (loads on primary truss) → Find $k_1$, $k_2$ (unit redundants) → Tabulate $\Sigma\,F_0\,kL$, $\Sigma\,k^2 L$

Solve for $X$ → Final $F = F_0 + kX$

### Example 5 — Three-bar concurrent truss (one degree, external/internal)




Three bars DA, DC, DB meeting at D; load W at D

- Vertical bar DC length $L$; inclined bars DA and DB at angle $\theta$ to the vertical, length $L/\cos\theta$. All have the same $AE$.
- Redundant: $X$ = force in DC. Equilibrium at D: $2F\cos\theta + X = W \rightarrow$ **$F = (W - X)/2\cos\theta$** ($F$ = force in each inclined bar).
- $U = X^2 L / 2AE + 2F^2 (L/\cos\theta)/2AE$; $\partial F/\partial X = -1/(2\cos\theta)$.
- $\partial U/\partial X = X L/AE + 2F(-1/2\cos\theta)(L/\cos\theta)/AE = 0 \rightarrow X = F/\cos^2\theta = (W - X)/(2\cos^3\theta)$.

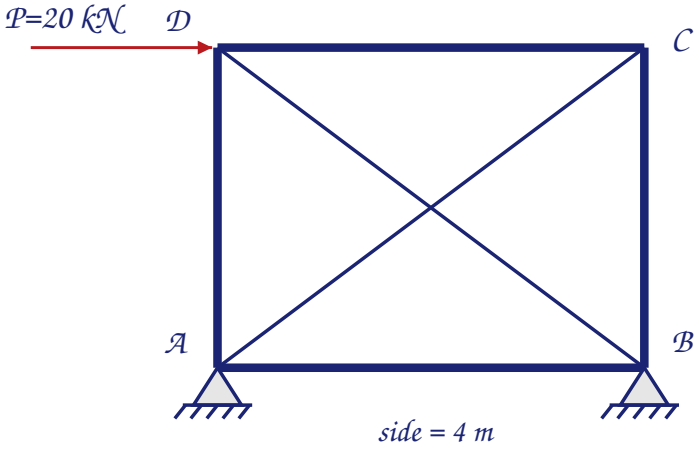
**Result**

**$X = W/(1 + 2\cos^3\theta)$** (force in the vertical bar, tension)

- Numerical: $\theta = 30°$, $W = 100$ kN $\rightarrow \cos^3 30° = 0.6495 \rightarrow$ **$X = 100/2.299 = 43.5$ kN**; $F = (100 - 43.5)/(2 \times 0.866) =$ **32.6 kN** in each inclined bar.
- Check: $2 \times 32.6 \times 0.866 + 43.5 = 100$ ✓.

## 8. Solution of Trusses with Two Redundants

### Example 6 — Square panel truss (side 4 m) with both diagonals, hinged at A and B




$m = 6, r = 4, j = 4$

$D_s = 6+4-8 = 2$

Square panel ABCD: both diagonals; pinned at A and B; load P = 20 kN at D

- Redundants: $X_1$ = horizontal reaction at B (external), $X_2$ = force in diagonal AC (internal, cut). Same $AE$ for all members.
- Primary truss: A hinged, B roller (vertical only), AC cut.

| Member | L (m) | $F_0$ (kN) | $k_1$ | $k_2$ |
|---|---|---|---|---|

| Member | $L$ (m) | $F_0$ (kN) | $k_1$ | $k_2$ |
|---|---|---|---|---|
| AB | 4 | +20 | +1 | −0.707 |
| BC | 4 | 0 | 0 | −0.707 |
| CD | 4 | 0 | 0 | −0.707 |
| DA | 4 | +20 | 0 | −0.707 |
| AC | 5.657 | 0 | 0 | +1 |
| BD | 5.657 | −28.28 | 0 | +1 |

- Sums: $\Sigma k_1^2 L = 4$; $\Sigma k_1 k_2 L = -2.83$; $\Sigma k_2^2 L = 19.31$; $\Sigma F_0 k_1 L = 80$; $\Sigma F_0 k_2 L = -273.14$.
- Equations ($AE$ cancels): $4 X_1 - 2.83 X_2 = -80$ and $-2.83 X_1 + 19.31 X_2 = +273.14$.
- Solving: $X_1 = -11.16$ kN ($H_B$ acts opposite to the assumed direction), $X_2 = +12.51$ kN (tension in AC).

| Member | Final force $F = F_0 + k_1 X_1 + k_2 X_2$ | Nature |
|---|---|---|
| AB | $20 - 11.16 - 8.85 = 0$ | zero force |
| BC | $-8.85$ | compression |
| CD | $-8.85$ | compression |
| DA | $20 - 8.85 = +11.16$ | tension |
| AC | $+12.51$ | tension |
| BD | $-28.28 + 12.51 = -15.78$ | compression |

- **Check joint D:** horizontal: $20 - 8.85 - 15.78\cos45° = 0$ ✓; vertical: $-11.16 + 15.78\sin45° = 0$ ✓.

> **Tip**
> Choose the redundants so that the primary truss is simple and as many $k$ values as possible are zero — it reduces the sums.

## 9. Indeterminate Trusses — Special Effects

- **Temperature rise $\Delta T$** in members: $\lambda = \alpha \Delta T L$; in a redundant truss it induces forces: $\Sigma (F_0 + k X) k L/AE + \Sigma k \alpha \Delta T L = 0$.
- **Lack of fit** (member too short/long by $e$): treat as an initial elongation $e$; the same type of term.
- **Support settlement $\delta_s$** at the redundant: $\partial U/\partial X = -\delta_s$ (sign by direction).
- **Self-stressing** truss: stresses appear even without external loads ($F_0 = 0$).
- **Effect of redundant member:** total strain energy is minimum; deflection is reduced.

## 10. Comparison — Castigliano I vs Castigliano II

| Basis | Castigliano's first theorem | Castigliano's second theorem |
|---|---|---|
| Statement | $\partial U/\partial P = \delta$ (deflection at $P$) | $\partial U/\partial R = 0$ (redundant $R$, unyielding support) |
| Used to find | deflection / slope | redundant reactions or member forces |
| Structure | determinate (or indeterminate once redundants are known) | indeterminate |
| Dummy load | needed if no load at the point | not needed; redundant is the variable |
| Final step | set dummy load to zero | solve the equation for $X$ |

## 11. Additional Solved Illustrations

### Illustration 1 — Propped cantilever with a point load at the middle (numerical)

- $L = 8$ m, $P = 48$ kN at the middle: $R_B = 5P/16 =$ **15 kN**; $R_A = 33$ kN; $M_A = -3PL/16 =$ **– 72 kN·m**; $M_C = 5PL/32 =$ **+ 60 kN·m**.

### Illustration 2 — Two-span beam, numerical

- Two equal spans of 5 m, UDL 24 kN/m: $R_B = 5wL/4 =$ **150 kN**; $R_A = R_C = 3wL/8 =$ **45 kN**; $M_B = -wL^2/8 =$ **– 75 kN·m**.

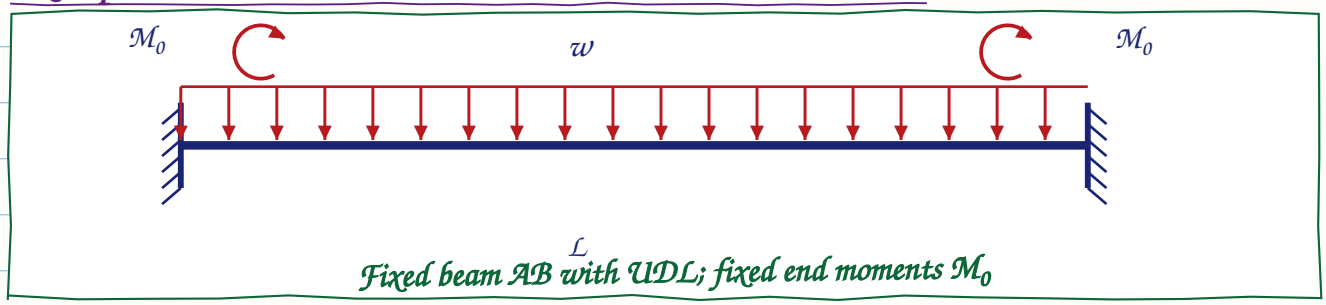
### Illustration 3 — Bars with a common joint (check)

- If $\theta = 0$, the three bars are all vertical: $X = W / 3$ — each carries equal load, as expected for equal $AE$.

## 11A. Fixed Beams by Castigliano II (link to Unit III)

- A fixed beam has $D_s = 3$; for vertical loads the horizontal reaction is zero, so take the **end moments** as redundants. By symmetry (symmetric loading) $M_A = M_B = M_0$.

### Example 7 — Fixed beam with UDL w over the whole span




Fixed beam AB with UDL; fixed end moments $M_0$

- Reactions $R_A = R_B = wL/2$. Taking sagging as positive: $M = wLx/2 - w x^2/2 - M_0$; $\partial M/\partial M_0 = -1$.
- $\partial U/\partial M_0 = 0$ (no rotation at the fixed end) $\rightarrow \int_0^L (wLx/2 - w x^2/2 - M_0)\, dx = 0$
- $\rightarrow w L^3/4 - w L^3/6 - M_0 L = 0 \rightarrow$ **$M_0 = w L^2 / 12$**.
- Centre moment = $wL^2/8 - wL^2/12 =$ **$+ wL^2/24$** (sagging). Points of contraflexure at 0.211 $L$

from each end.

### Example 8 — Fixed beam with central point load P

- For $0 \le x \le L/2$: $M = P x/2 - M_0$. By symmetry integrate over half: $\int_0^{L/2} (P x/2 - M_0)\, dx = 0$
- $\rightarrow P L^2/16 - M_0 L/2 = 0 \rightarrow \mathbf{M_0 = P L / 8}$ ; centre moment $= PL/4 - PL/8 = \mathbf{+ PL/8}$ .
- These fixed-end moments are the basis of **slope-deflection and moment distribution** (Units IV and V).

## 11B. Cross-Check by Consistent Deformation (Compatibility Method)

- Same propped cantilever with UDL: remove the prop → free cantilever; deflection at B due to UDL = $w L^4 / 8EI$ (downward).
- Apply upward R at B → deflection = $R L^3 / 3EI$ (upward).
- Compatibility: $\mathbf{w L^4 / 8EI = R L^3 / 3EI \rightarrow R = 3wL/8}$ √ (same as Castigliano II).
- Two-span beam check: deflection at B of a simply supported beam of span 2L with UDL = 5 w $(2L)^4 / 384EI = 5wL^4/24EI$; due to R: $R(2L)^3/48EI = RL^3/6EI$; equate → $\mathbf{R = 5wL/4}$ √.

| Basis | Castigliano II | Consistent deformation |
|---|---|---|
| Idea | minimum strain energy | geometric compatibility at the redundant |
| Works with | M or F expressions in terms of X | standard deflection formulae |
| Best for | trusses, frames, curved members | beams with standard cases |
| Computation | integration needed | uses tabulated results — fast |

## 11C. Practice — Find $D_s$ and $D_k$ (with answers)

| No. | Structure | $D_s$ | $D_k$ |
|---|---|---|---|
| 1 | Beam fixed at A, rollers at B and C (two spans) | 5 − 3 = 2 | rotations at B, C = 2 |
| 2 | Plane truss: j = 6, m = 10, r = 3 | 10 + 3 − 12 = 1 | 2j − r = 9 |
| 3 | Fixed-fixed beam with an internal hinge | 6 − 3 − 1 = 2 | rotation on both sides of the hinge + deflection = 2 |
| 4 | Single-bay, two-storey frame, fixed bases (6 joints, 6 members, r = 6) | 18 + 6 − 18 = 6 | 4 rotations + 2 sways = 6 |
| 5 | Propped cantilever | 1 | 1 |
| 6 | Hinged-base portal frame | 1 | 5 |

| No. | Structure | $D_s$ | $D_k$ |
|---|---|---|---|
| 7 | Plane truss: $j = 7$, $m = 11$, $r = 3$ (simply supported) | $11 + 3 - 14 = 0$ (determinate) | $14 - 3 = 11$ |
| 8 | Closed rectangular ring (box) | 3 | – |

→ $D_k$ is different if axial deformation is considered or neglected — mention your assumption in the answer.

## 11D. Settlement, Temperature and Redundant Structures

- **Determinate structure:** no stresses due to settlement or temperature (it just moves).
- **Indeterminate structure:** settlement or temperature change **induces** stresses and reactions — a drawback, but also useful (prestressing by jacking supports).
- Fixed beam, one end sinks $\Delta$: end moments $\propto 6EI\Delta/L^2$ (see Unit III).
- A truss with a temperature rise in one member causes force in the redundant members.

## 11E. Stability, Mechanism and Geometry (supplement)

- Equation $D_s \geq 0$ is **necessary but not sufficient** . Check geometry: the reactions must not be concurrent or all parallel; no part should be free to move (mechanism).
- Example of unstable arrangement: a beam on three **roller** supports (all vertical) has $r = 3$ but cannot resist horizontal load → **unstable** .
- A truss with a **non-triangulated** panel is a mechanism (needs a diagonal).
- Check: **$D_s = 0$ and stable** → determinate; **$D_s > 0$ and stable** → indeterminate; **$D_s < 0$** → unstable.

## 12. Quick Revision — Unit II

### Standard results

Truss: $D_s = m + r - 2j$, $D_k = 2j - r$ | Frame: $D_s = 3m + r - 3j$, $D_k = 3j - r$ | Beam: $D_s = r - 3$ | Castigliano II: $\partial U/\partial R = 0$ | $F = F_0 + kX$ | $X = -\Sigma F_0 kL \div \Sigma k^2 L$.

Propped cantilever UDL: $R_B = 3wL/8$, $M_A = -wL^2/8$ | Point load at centre: $R_B = 5P/16$ | Two-span continuous beam UDL: $R_B = 5wL/4$.

### Possible exam questions

1. Differentiate between determinate and indeterminate structures. Explain static and kinematic indeterminacy.
2. Determine $D_s$ and $D_k$ for the given frames, beams and trusses.
3. State and prove Castigliano's second theorem (least work).
4. Analyse a propped cantilever/continuous beam using Castigliano's second theorem.

**5.** Analyse the indeterminate truss with one redundant member; two redundants (numerical).

# UNIT - III

## Fixed Beams and Continuous Beams

## 1. Introduction to Statically Indeterminate Beams

- Beams with **more than three reaction components** cannot be solved by statics alone — they are **statically indeterminate beams** .
- Main types: **(i) propped cantilever, (ii) fixed (built-in) beam, (iii) continuous beam** .




Types of indeterminate beams

- Methods: **Castigliano II, consistent deformation, moment-area, three-moment theorem (Clapeyron), slope-deflection, moment distribution** .

## 2. Fixed Beams — Meaning and Features

- A **fixed beam (built-in or encastré beam)** has both ends rigidly fixed so that **slope and deflection at both ends are zero** .
- Reaction components: at each end vertical reaction, horizontal reaction, and **fixed end moment** → 6 unknowns; $D_s = 3$ ( **2 for vertical loads** since $H = 0$).

| Basis | Simply supported beam | Fixed beam |
|---|---|---|
| End conditions | free to rotate; $M = 0$ at ends | end rotation prevented; fixed end moments $M_A$, $M_B$ |
| Max bending moment | $wL^2/8$ (UDL) | $wL^2/12$ (at supports) — **33% less** |
| Max deflection (UDL) | $5wL^4/384EI$ | $wL^4/384EI$ — **1/5th** |
| Stiffness | less | more |
| Economy | larger sections | smaller sections |
| Settlement of support | no stresses | causes stresses (disadvantage) |
| Analysis | simple (equilibrium) | indeterminate |

### Advantages of fixed beams

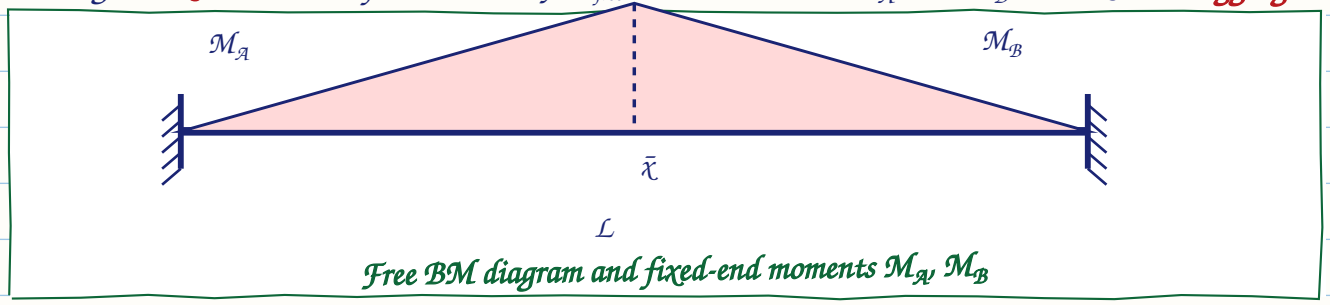
- Smaller maximum bending moment and deflection; more stiff and strong; economical section; more stable under load.

### Disadvantages

- Fixed end moments need heavy end connections; **settlement, temperature** and rotation of supports produce extra stresses; analysis is difficult; fixed ends are hard to achieve in practice.

## 3. Fixed End Moments — Area-Moment Method

- Consider fixed beam AB, span $L$, constant $EI$. Let **$a$** = area of the free (simply supported) BM diagram, **$\bar{x}$** = distance of its centroid from A. End moments $M_A$ and $M_B$ are taken as **hogging**.




Free BM diagram and fixed-end moments $M_A$, $M_B$

- **Conditions** (slopes at A and B are the same, so change of slope = 0; deflection of B from tangent at A = 0):

1. Area of net BM diagram = 0: $a = (M_A + M_B)\,L/2$.
2. Moment of net M diagram about A = 0: $a\bar{x} = M_A L^2/6 + M_B L^2/3$ (moment of the trapezoid about A). Solving both conditions:

**Fixed end moments (general)**

$$M_A = 2a(2L - 3\bar{x})/L^2 \quad M_B = 2a(3\bar{x} - L)/L^2$$

- Check for central load: $a = PL^2/8$, $\bar{x} = L/2 \rightarrow M_A = M_B = PL/8$ ✓.
- **Reactions:** $R_A$ = (free reaction at A) + $(M_A - M_B)/L$ ; $R_B$ = (free reaction at B) − $(M_A - M_B)/L$.
- **Net BM at any section = free BM − fixed end moment diagram (trapezoid)**.

### Procedure

Draw free BM diagram → Find area a and centroid $\bar{x}$ → Find $M_A$ and $M_B$ → Find reactions

Draw SFD and BMD → Locate max BM and contraflexure

→ If end moments are equal (symmetrical loading) $M_A = M_B = a/L$ (average height of the free BM diagram).

## 4. Fixed End Moments for Standard Cases

| Loading on fixed beam of span $L$ | $M_A$ (fixed end A) | $M_B$ (fixed end B) |
|---|---|---|
| UDL $w$ over the whole span | $wL^2/12$ | $wL^2/12$ |
| Central point load $P$ | $PL/8$ | $PL/8$ |
| Eccentric point load $P$ at $a$ from A, $b$ from B | $Pab^2/L^2$ | $Pa^2b/L^2$ |
| Two equal loads $P$ at $L/3$ and $2L/3$ | $2PL/9$ | $2PL/9$ |
| Triangular load, zero at A to $w$ at B (UVL) | $wL^2/30$ | $wL^2/20$ |
| Triangular load, $w$ at A to zero at B | $wL^2/20$ | $wL^2/30$ |
| Triangular (symmetrical, peak $w$ at centre) | $5wL^2/96$ | $5wL^2/96$ |
| UDL $w$ over left half only (length $L/2$ from A) | $11wL^2/192$ | $5wL^2/192$ |
| Couple $M_0$ at distance $a$ from A ($b = L - a$) | $M_0 b(2a - b)/L^2$ | $M_0 a(2b - a)/L^2$ |

- For downward loads the end moments are **hogging**. For the couple the magnitudes are given (a clockwise couple is taken positive); one end is hogging and the other sagging.

### Derivation 1 — UDL (a quick proof)

- Free BM $= wLx/2 - wx^2/2$, area $a = (2/3)\cdot L\cdot(wL^2/8) = wL^3/12$, $\bar{x} = L/2 \rightarrow M_A = 2a(2L - 1.5L)/L^2 =$ **$wL^2/12$**.

### Derivation 2 — Eccentric point load

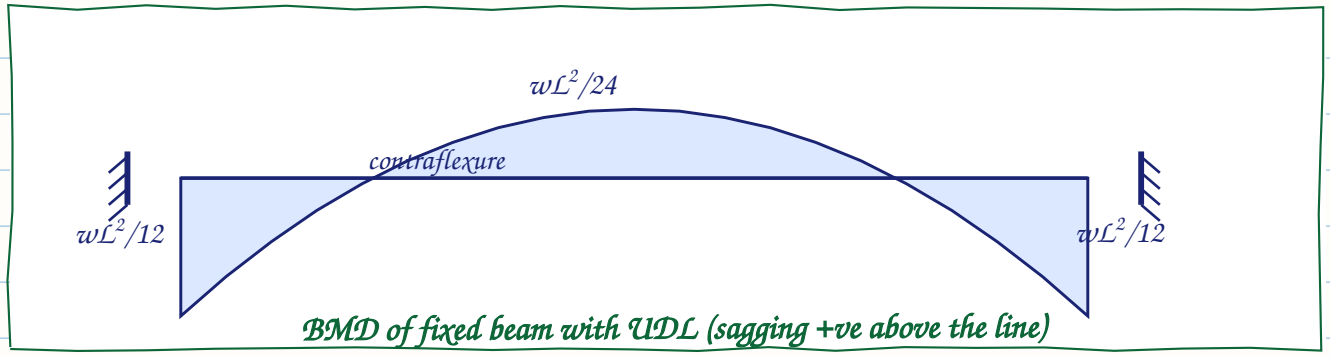
- Free BM is a triangle with peak $Pab/L$ under the load: area $= \tfrac{1}{2} \times L \times (Pab/L) = Pab/2$; centroid from A: $\bar{x} = (L + a)/3$.
- $M_A = 2\,(Pab/2)(2L - (L + a))/L^2 =$ **$Pab^2/L^2$** and likewise $M_B =$ **$Pa^2b/L^2$** (the larger moment is at the end nearer to the load).

### Derivation 3 — UVL (zero at A, $w$ at B)

- $R_{A0} = wL/6$; $M(x) = wLx/6 - wx^3/6L$; area $a = wL^3/24$; $\bar{x} = 8L/15 \rightarrow M_A =$ **$wL^2/30$**, $M_B =$ **$wL^2/20$**.

## 5. Fixed Beam — Reactions, SFD and BMD (Worked Examples)

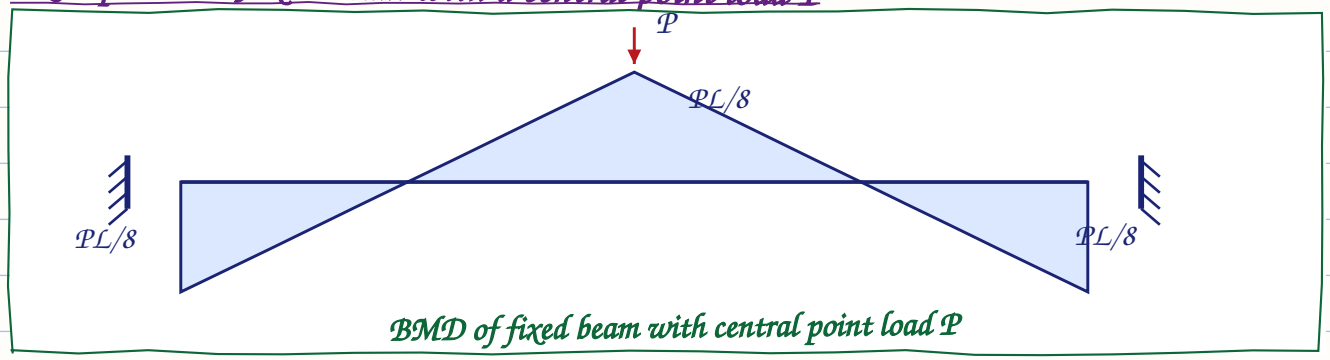
### Example 1 — Fixed beam with UDL over the whole span




BMD of fixed beam with UDL (sagging +ve above the line)

- $M_A = M_B = wL^2/12$ (hogging); $R_A = R_B = wL/2$. Centre BM = $wL^2/8 - wL^2/12 =$ **$+ wL^2/24$** .
- **Points of contraflexure** at $x = L(1 \pm 1/\sqrt{3})/2 =$ **0.211 L** from each end.
- SFD: straight line from $+ wL/2$ at A to $- wL/2$ at B, zero at the centre.

## Example 2 — Fixed beam with a central point load P




BMD of fixed beam with central point load P

- $M_A = M_B = PL/8$; $R_A = R_B = P/2$; net M at the centre = $PL/4 - PL/8 =$ **$+ PL/8$** . Contraflexure at $L/4$ from each end.
- SFD: $+ P/2$ constant in AC and $- P/2$ in CB (a jump of P at the load).

## Example 3 — Numerical: fixed beam with UDL + central point load

**Problem**

A fixed beam of span 6 m carries a UDL of 20 kN/m over the entire span and a point load of 40 kN at the centre. Find the fixed end moments, reactions and the centre BM.

- UDL: $wL^2/12 = 20 \times 36/12 = 60$ kN·m; point load: $PL/8 = 40 \times 6/8 = 30$ kN·m.
- **$M_A = M_B = 60 + 30 = 90$ kN·m** (hogging).
- **$R_A = R_B = (20 \times 6 + 40)/2 = 80$ kN**.
- Free BM at centre = $wL^2/8 + PL/4 = 90 + 60 = 150$ kN·m → **net centre BM = 150 − 90 = 60 kN·m** (sagging).

## Example 4 — Eccentric point load (numerical)

**Problem**

Fixed beam of span 6 m; P = 60 kN at 2 m from A. Find $M_A$, $M_B$, $R_A$, $R_B$ and BM under the load.

- $a = 2$ m, $b = 4$ m: $M_A = P\,a\,b^2/L^2 = 60 \times 2 \times 16 / 36 =$ **53.33 kN·m** ; $M_B = P\,a^2\,b/L^2 = 60 \times 4 \times 4 / 36 =$ **26.67 kN·m** .

- $R_A = P\,b^2\,(3a + b)/L^3 = 60 \times 16 \times 10 / 216 =$ **44.44 kN**; $R_B = 60 - 44.44 =$ **15.56 kN**.
- $BM$ under load $= R_A \times 2 - M_A = 88.89 - 53.33 =$ **+ 35.56 kN·m** (check from right: $15.56 \times 4 - 26.67 = 35.56$ √).

### Example 5 — Number of point loads and combination

- Use **superposition**: sum the fixed end moments of each load separately, e.g., loads of 30 kN at 2 m and 20 kN at 4 m on a 6 m beam.
- Case 30 kN: $a = 2, b = 4 \rightarrow M_A = 30 \times 2 \times 16/36 = 26.67$; $M_B = 30 \times 4 \times 4/36 = 13.33$.
- Case 20 kN: $a = 4, b = 2 \rightarrow M_A = 20 \times 4 \times 4/36 = 8.89$; $M_B = 20 \times 16 \times 2/36 = 17.78$.
- **Total $M_A = 35.56$ kN·m, $M_B = 31.11$ kN·m**. free $R_{A0} = (30 \times 4 + 20 \times 2)/6 = 26.67$ kN; $R_A = R_{A0} + (M_A - M_B)/L = 26.67 + (35.56 - 31.11)/6 =$ **27.41 kN**; $R_B = 50 - 27.41 =$ **22.59 kN**.

### Example 6 — Fixed beam with UVL (numerical)

- Span 6 m, load rising from 0 at A to 24 kN/m at B: $M_A = wL^2/30 = 24 \times 36/30 =$ **28.8 kN·m**; $M_B = wL^2/20 =$ **43.2 kN·m**.
- Total load $= ½ \times 24 \times 6 = 72$ kN; free $R_{A0} = wL/6 = 24$ kN; $R_A = 24 + (28.8 - 43.2)/6 =$ **21.6 kN**; $R_B = 72 - 21.6 =$ **50.4 kN**.

### Example 7 — Fixed beam with a couple $M_0$

- Couple $M_0$ at the centre: $M_A = M_B = M_0/4$ (equal in magnitude — one end hogging, the other sagging); reactions $R_A = -R_B = 3M_0/2L$ (a couple of reactions).
- For a couple at a distance $a$ from A: $M_A = M_0\,b\,(2a - b)/L^2$, $M_B = M_0\,a\,(2b - a)/L^2$ (clockwise couple positive).

## 6. Deflection of Fixed Beams

- Fixed beams are much stiffer. Deflection is obtained by the **net BM diagram** (moment-area) or **superposition** of the simple-beam deflection and the deflection due to end moments.

| Loading | Max deflection (at centre) | Compare S.S. |
|---|---|---|
| UDL $w$ over full span | $w\,L^4 / 384\,EI$ | $5wL^4/384EI$ (5 times) |
| Central point load $P$ | $P\,L^3 / 192\,EI$ | $PL^3/48EI$ (4 times) |
| Eccentric load $P$ ($a$ from A, $b$ from B) | under the load: $P\,a^3\,b^3 / 3\,E\,I\,L^3$ | — |

- **Derivation (UDL):** $\delta_{centre} = \delta$(simple) $- \delta$(due to end moments $M_0 = wL^2/12$) $= 5wL^4/384EI - (M_0\,L^2/8EI) = wL^4/EI\,(5/384 - 4/384) =$ **$wL^4/384EI$**.
- For an eccentric load the maximum deflection occurs in the **longer segment** (where the slope is zero); it is slightly more than the deflection under the load.

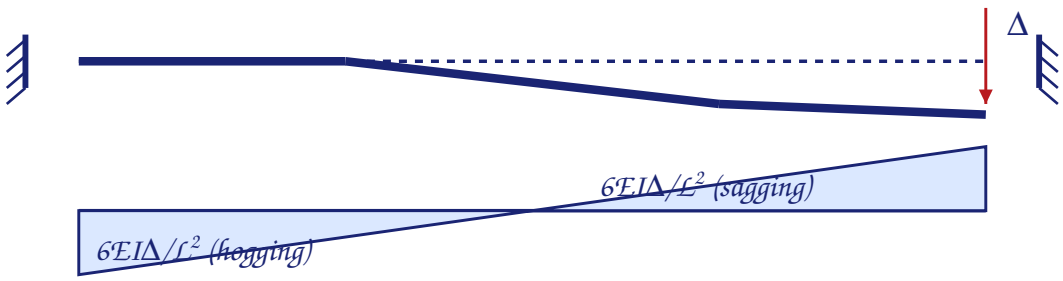
## 7. Effect of Sinking (Settlement) of Support

- If one support B of a fixed beam **sinks by Δ** (no rotation of ends), the beam deforms like a guided bar; **no load needed** — moments and shears arise from the displacement.
- Chord rotation $\psi = \Delta/L$.

**Fixed end moments due to settlement Δ**

$M_A = M_B = 6EI\Delta / L^2$ (A hogging, B sagging) (BM varies linearly)

Shear $= 12EI\Delta / L^3$ (constant) — $R_A$ increases, $R_B$ decreases by this amount




Fixed beam, support B sinks by Δ : beam shape and BMD

- **Derivation:** Slope-deflection: $M_{AB} = 2EI/L\,(2\theta_A + \theta_B - 3\psi)$ with $\theta_A = \theta_B = 0 \rightarrow M_{AB} = -6EI\Delta/L^2$ ; similarly $M_{BA} = -6EI\Delta/L^2$.
- The final moments = moments due to loads + moments due to settlement (superposition).

### Example 8 — Numerical (settlement)

**Problem**

Fixed beam, span 6 m, $EI = 30000$ kN·m², right support sinks by 10 mm. Find end moments and reaction change.

- $M = 6EI\Delta/L^2 = 6 \times 30000 \times 0.01 / 36 =$ **50 kN·m** .
- $\Delta R = 12EI\Delta/L^3 = 12 \times 30000 \times 0.01 / 216 =$ **16.67 kN**.
- If a UDL $w = 20$ kN/m also acts: $M_A = wL^2/12 + 50 = 60 + 50 = 110$ kN·m (hogging at A); $M_B = 60 - 50 = 10$ kN·m (hogging at B).

### Observation

- Settlement **increases** the moment at one end and **reduces** it at the other — a serious point in design of indeterminate beams.

## 8. Effect of Rotation of a Support

- If the end A rotates by θ (clockwise) with B still fixed and no relative deflection:

**Moments due to rotation θ of support A**

$M_A = 4EI\theta / L$ , $M_B = 2EI\theta / L$ (carry-over factor ½)

Shear = $6EI\theta / L^2$ ($R_A$ and $R_B$ equal and opposite)

- If **both ends rotate** by $\theta_A$ and $\theta_B$: $M_A = (2EI/L)(2\theta_A + \theta_B)$, $M_B = (2EI/L)(2\theta_B + \theta_A)$.
- **Example:** $L = 5$ m, $EI = 20000$ kN·m², end A rotates 0.002 rad: $M_A = 4 \times 20000 \times 0.002 / 5 =$ **32 kN·m** ; $M_B =$ **16 kN·m** ; shear $= 6 \times 20000 \times 0.002/25 =$ **9.6 kN**.
- A simply supported (pin) end has **zero** moment after rotation; so rotation of a support matters only for a **fixed** end.

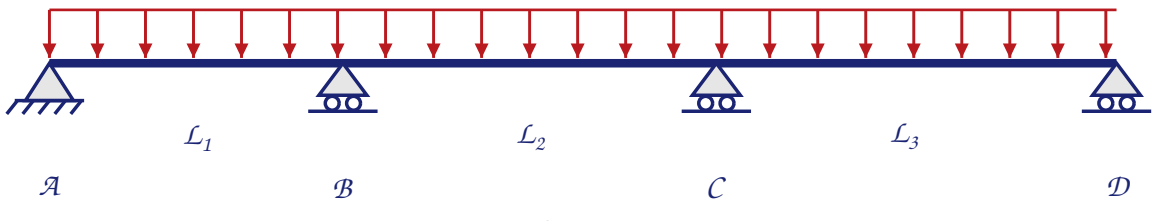
## 9. Propped Cantilever — Standard Results

| Loading (A fixed, B propped) | $R_B$ | $M_A$ |
|---|---|---|
| UDL w over full span | $3wL/8$ | $wL^2/8$ (hogging) |
| Central point load P | $5P/16$ | $3PL/16$ |
| Point load P at a from A | $P\,a^2\,(3L - a) / 2L^3$ | $P\,a\,b\,(L + b) / 2L^2$ |

- Prop reaction is found from **zero deflection at the prop** (Unit II methods).

## 10. Continuous Beams — Introduction

- A **continuous beam** rests on **more than two supports** ; it is continuous over the intermediate supports.
- Degree of indeterminacy = (number of reaction components) − 3 = **n − 2** for n simple supports.




Continuous beam with three spans

### Advantages

- Smaller BM than a series of S.S. beams; economical; fewer expansion joints; more rigid; smooth riding (bridges); redistribution of moments on overload.

### Disadvantages

- Analysis is difficult; **settlement of supports** changes the moments; **temperature** ; and a hogging moment at the supports needs reinforcement on the top.

### Features of BMD

- **Hogging** BM over the interior supports and **sagging** BM in the middle of the spans; points of contraflexure exist in every span.

# 11. Clapeyron's Theorem of Three Moments

- Gives a relation between the **hogging moments at three consecutive supports**. Each span has its own free BM diagram.

**Theorem of three moments (supports at same level, EI constant, hogging +ve)**

$$M_A L_1 + 2 M_B (L_1 + L_2) + M_C L_2 = 6 a_1 \bar{x}_1 / L_1 + 6 a_2 \bar{x}_2 / L_2$$

- $a_1$, $a_2$ = areas of free BM diagrams of spans AB and BC; $\bar{x}_1$ = distance of centroid of $a_1$ from **A** (the left support); $\bar{x}_2$ = distance of centroid of $a_2$ from **C** (the right support).
- **Standard right-hand sides ($6 a \bar{x} / L$):**

| Load on a span of length $L$ | $6 a \bar{x} / L$ |
|---|---|
| UDL $w$ | $w L^3 / 4$ |
| Central point load $P$ | $3 P L^2 / 8$ |
| Point load $P$ at $a$ from the end considered ($b = L - a$) | $P a b (L + a)/L$ (from the left end); $P a b (L + b)/L$ (from the right end) |
| UVL (zero at one end, $w$ at the other) | $2 w L^3/15$ ($\bar{x}$ measured from the zero end); $7 w L^3/60$ ($\bar{x}$ measured from the max end) |

## With support settlement

- If B is lower than A and C by $\delta_1$ and $\delta_2$: $M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = 6a_{1\bar{x}1}/L_1 + 6a_{2\bar{x}2}/L_2 - 6EI\,(\delta_1/L_1 + \delta_2/L_2)$.
- **Fixed end A:** add an imaginary span of zero length → $2 M_A L_1 + M_B L_1 = 6 a_1 \bar{x}_1 / L_1$.
- **Overhang:** the moment at the support = (load × distance) known; use it as $M_A$ in the equation.

## Procedure

Draw free BM diagrams → Write 3-moment equation for every pair of adjacent spans → Solve for support moments → Reactions

SFD and BMD

1. Number of three-moment equations = number of unknown support moments (= $D_s$).
2. Solve simultaneous equations; find reactions of every span: $R$ = (free reaction) ± $(M_B - M_A)/L$.
3. Draw SFD, then BMD (free BM + support-moment trapezoid).

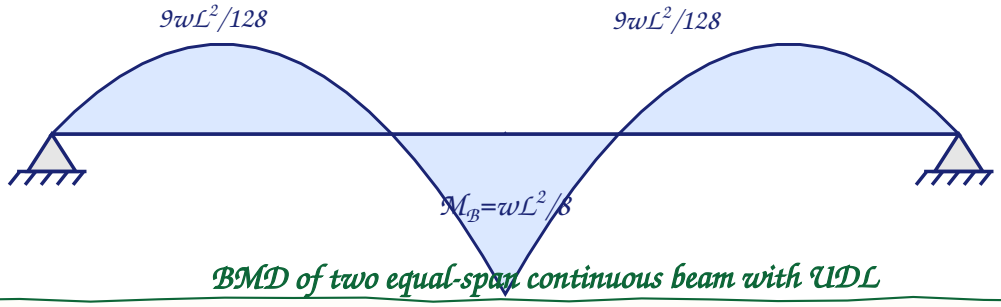
## 12. Continuous Beam — Solved Examples

### Example 9 — Two spans with UDL

**Problem**

Continuous beam ABC, simply supported at A and C: AB = 4 m, BC = 6 m, UDL 20 kN/m on the whole length. Find the support moment and reactions.

- $M_A = M_C = 0$. Equation: $2\,M_B\,(4 + 6) = w\,(L_1^3 + L_2^3)/4 = 20\,(64 + 216)/4 = 1400 \rightarrow$ **$M_B$ = 70 kN·m (hogging)**.
- $R_A = wL_1/2 - M_B/L_1 = 40 - 17.5 =$ **22.5 kN**; $R_C = 60 - 70/6 =$ **48.33 kN**; **$R_B$ = 200 − 22.5 − 48.33 = 129.17 kN**.
- Max sagging in AB: $x = 22.5/20 = 1.125$ m from A → $M = 22.5 \times 1.125 - 10 \times 1.125^2 =$ **12.66 kN·m**.
- Max sagging in BC: $x = 48.33/20 = 2.417$ m from C → $M = 48.33 \times 2.417 - 10 \times 2.417^2 =$ **58.4 kN·m**.




BMD of two equal-span continuous beam with UDL

- The figure shows the equal-span case: $M_B = wL^2/8$ and sagging $9wL^2/128$.

### Example 10 — Two equal spans with central point loads

- $L = 6$ m each, $P = 60$ kN at the centre of each span: $4\,M_B\,L = 2 \times 3PL^2/8 \rightarrow$ **$M_B = 3PL/16 = 3 \times 60 \times 6/16 = 67.5$ kN·m**.
- $R_A = P/2 - M_B/L = 30 - 11.25 =$ **18.75 kN** $= R_C$; $R_B = 120 - 37.5 =$ **82.5 kN**.
- BM under each load = $18.75 \times 3 =$ **56.25 kN·m** (sagging).

### Example 11 — Beam fixed at A (two spans)

**Problem**

Beam ABC: A fixed, B and C rollers; AB = BC = 6 m; UDL 20 kN/m on both spans. Find the support moments and reactions.

- Fixed end A (imaginary zero span): $2\,M_A\,(6) + M_B\,(6) = w\,L^3/4 = 1080 \rightarrow$ **$2\,M_A + M_B = 180$**.
- Equation at B ($M_C = 0$): $6\,M_A + 24\,M_B = 1080 + 1080 \rightarrow$ **$M_A + 4\,M_B = 360$**.
- Solving: **$M_A$ = 51.43 kN·m, $M_B$ = 77.14 kN·m** (both hogging).
- Span AB: $R_A = 60 + (M_A - M_B)/6 =$ **55.71 kN**; $R_B$(left) = 64.29 kN. Span BC: $R_C = 60 -$

$77.14/6 = \mathbf{47.14\ kN}$; $R_B$(right) = 72.86 kN.

- $\mathbf{R_B = 137.14\ kN}$; check: 55.71 + 137.14 + 47.14 = 240 = total load √.

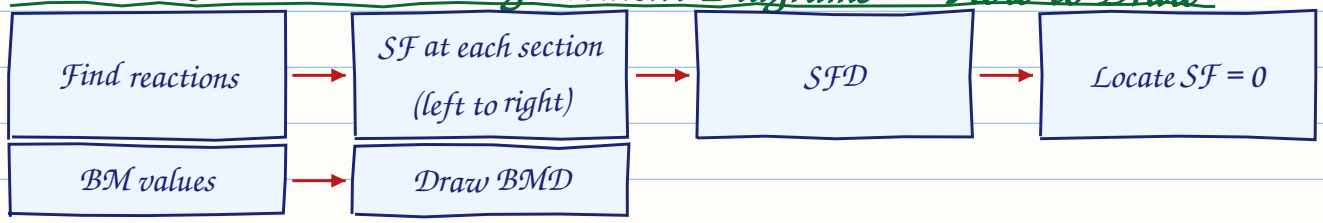
### Example 12 — Effect of settlement

**Problem**

Two equal spans $L$ = 6 m, UDL $w$ = 20 kN/m, $EI$ = 30000 kN·m²; the middle support B settles by 10 mm. Find $M_B$.

- $4\,M_B\,L = w\,L^3/2 - 6EI\Delta\,(2/L) \rightarrow M_B = wL^2/8 - 3EI\Delta/L^2 = 90 - 25 = \mathbf{65\ kN{\cdot}m}$.
- Without settlement $M_B$ = 90 kN·m; settlement **reduces** the hogging moment at B (and increases the sagging in the spans).

## 13. Shear Force and Bending Moment Diagrams — How to Draw




- **SFD:** starts with $R_A$, drops by load, jumps up at a reaction; slope = − $w$ for UDL.
- **BMD:** area under SFD gives change in BM; **BM is max where SF = 0**.
- **Points of contraflexure:** where BM changes sign (BM = 0).
- For fixed beams: BMD = free BM diagram (parabola / triangle) **minus** the end-moment trapezoid; for continuous beams = free BM + support-moment line.

### Sign conventions

- Sagging BM +ve (draw above/below as per your textbook — mention clearly); hogging −ve.
- SF +ve when the left part tends to move up relative to the right.

## 14. Comparison — Fixed vs Continuous vs Propped Cantilever

| Basis | Propped cantilever | Fixed beam | Continuous beam |
|---|---|---|---|
| Supports | fixed + roller | fixed + fixed | three or more supports |
| $D_s$ | 1 | 3 (2) | $n - 2$ |
| UDL max BM | $wL^2/8$ | $wL^2/12$ | $wL^2/8$ at support (equal spans) |
| Best method | Castigliano II | area-moment | three-moment, slope deflection |

## 14A. Three-Span Continuous Beam — Equal Spans with UDL

- Supports A, B, C, D; spans $L$ each; symmetry gives $M_B = M_C = M$; $M_A = M_D = 0$.
- Equation at B: $2M(2L) + ML = 2 \times wL^3/4 \rightarrow 5ML = wL^3/2 \rightarrow$ **$M_B = M_C = wL^2/10$** .
- Numerical: $L = 5$ m, $w = 20$ kN/m → **$M = 20 \times 25/10 = 50$ kN·m (hogging)** .
- $R_A = wL/2 - M/L = 50 - 10 =$ **40 kN** $= R_D$; $R_B = R_C = (300 - 80)/2 =$ **110 kN** $(= 1.1\,wL)$.
- Sagging BM: end span $x = 40/20 = 2$ m → $40 \times 2 - 10 \times 4 =$ **40 kN·m** ; middle span centre: $wL^2/8 - M = 62.5 - 50 =$ **12.5 kN·m** .

| Equal spans, UDL | Two-span | Three-span | S.S. span |
|---|---|---|---|
| Support moment | $wL^2/8$ | $wL^2/10$ | 0 |
| End-support reaction | $3wL/8$ | $2wL/5$ | $wL/2$ |
| Interior reaction | $5wL/4$ | $11wL/10$ | — |

## 14B. Continuous Beam with Overhang and Mixed Loads

**Problem**

Beam on supports A, B, C with spans AB = BC = 6 m and an overhang CD = 2 m carrying a point load of 30 kN at D. A UDL of 10 kN/m acts on AB and BC. Find the support moments and reactions.

- **Overhang moment at C** (known from statics): $M_C = -30 \times 2 = -60$ kN·m, i.e. **hogging 60 kN·m** .
- Equation at B: $M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = wL_1^3/4 + wL_2^3/4 \rightarrow 0 + 24M_B + 360 = 540 + 540 \rightarrow$ **$M_B = (1080 - 360)/24 = 30$ kN·m** (hogging).
- Span AB: $R_A = 30 + (0 - 30)/6 =$ **25 kN**; $R_B$(left) $= 30 + 5 = 35$ kN.
- Span BC (hogging 30 at B and 60 at C): $R_B$(right) $= 30 + (30 - 60)/6 = 25$ kN; $R_C$(span) $= 30 + 5 = 35$ kN.
- **$R_B = 35 + 25 = 60$ kN**; **$R_C = 35 + 30$ (overhang load) $= 65$ kN**; check: $25 + 60 + 65 = 150 = 120 + 30$ √.

→ Treat the overhang as a cantilever: its moment at the support is fixed by statics and used in the equation.

## 14C. Propped Cantilever — More Cases (A fixed, B propped)

- **Couple $M_0$ applied at the prop B:** $R_B = 3M_0/2L$, $M_A = M_0/2$ (carry-over).
- **UDL over the whole span:** $R_B = 3wL/8$, $M_A = wL^2/8$ (hogging); BM at $3L/8 = 9wL^2/128$.
- **Point load at a from fixed end A:** $R_B = P\,a^2\,(3L - a)/2L^3$.
- **Check with $a = L/2$:** $R_B = 5P/16$ √.
- Example: $L = 6$ m, $P = 48$ kN at 2 m from A: $R_B = 48 \times 4 \times (18 - 2)/(2 \times 216) =$ **7.11 kN**; $R_A$

$= 48 - 7.11 = 40.89$ kN; $M_A = P\,a - R_B\,L = 96 - 42.67 =$ **53.33 kN·m** (hogging). Check: $P\,a\,b\,(L + b)/2L^2 = 48 \times 2 \times 4 \times 10/72 = 53.33$ √.

## 14D. Redistribution of Moments in Continuous Beams

- In reinforced concrete and steel design, continuous beams develop **plastic hinges** at supports on overload; the BM diagram then **redistributes** — moments at supports reduce and span moments increase.
- This is why indeterminate structures have **reserve strength**; codes allow limited (up to 15–30%) redistribution.
- Elastic analysis (three moments, slope-deflection, moment distribution) is still the starting point.

## 14E. Common Mistakes in Fixed and Continuous Beam Problems

- Confusing **a** (area of free BM) and $\bar{x}$ (distance from A) — recheck with a central load where $M = PL/8$.
- Using wrong sign for settlement — settlement of the middle support **reduces** hogging at that support.
- For the three-moment theorem: **$\bar{x}$ is measured from the outer end** (left end of span 1, right end of span 2).
- Forgetting the fixed-end condition (imaginary zero-length span).
- Not adding the **overhang moment** at the support.
- Drawing the BMD without marking the **maximum moments and points of contraflexure**.
- Not checking $\Sigma$ reactions = total load.

## 14F. Practice Questions with Answers

| No. | Problem | Answer |
|---|---|---|
| 1 | Fixed beam 8 m, UDL 15 kN/m | $M_A = M_B = 80$ kN·m; centre +40 kN·m |
| 2 | Fixed beam 5 m, central load 40 kN | $M_A = M_B = 25$ kN·m; centre +25 kN·m |
| 3 | Fixed beam 6 m, load 36 kN at 2 m from A | $M_A = 32$ kN·m, $M_B = 16$ kN·m |
| 4 | Continuous beam, 2 equal spans of 4 m, UDL 25 kN/m | $M_B = 50$ kN·m; $R_B = 125$ kN |
| 5 | Fixed beam 4 m, EI = 20000, end sinks 5 mm | $M = 6EI\Delta/L^2 = 37.5$ kN·m |
| 6 | Fixed beam 6 m, UVL 0 to 30 kN/m (A to B) | $M_A = 36$ kN·m; $M_B = 54$ kN·m |

## 15. Quick Revision — Unit III

### Formulae

Fixed beam: $M_A = 2a(2L - 3\bar{x})/L^2$, $M_B = 2a(3\bar{x} - L)/L^2$ | UDL: $wL^2/12$ | Central load: $PL/8$ | Eccentric: $Pab^2/L^2$, $Pa^2b/L^2$ | Settlement: $6EI\Delta/L^2$ | Rotation $\theta$: $4EI\theta/L$, $2EI\theta/L$ | Three moments: $M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = 6a_{1}\bar{x}_1/L_1 + 6a_{2}\bar{x}_2/L_2$ | Equal spans UDL: $M_B = wL^2/8$, $R_B = 5wL/4$.

### Possible exam questions

1. Derive the fixed end moments for a fixed beam of span $L$ with an eccentric point load.
2. Draw SFD and BMD for a fixed beam with UDL/central load; locate points of contraflexure.
3. Explain the effect of sinking of a support on the fixed end moments (derive).
4. State and derive Clapeyron's theorem of three moments.
5. Analyse a two-span continuous beam with UDL and point loads; draw SFD and BMD.
6. A continuous beam with a settlement of support (numerical).

# UNIT - IV

## Slope-Deflection Method

## 1. Introduction

- The **slope-deflection method** was introduced by **G. A. Maney (1915)**. It is a **displacement (stiffness) method**: the **joint rotations and joint translations** are the unknowns (kinematic indeterminacy $D_k$).
- The end moments of every member are written in terms of the **joint displacements** (slope $\theta$ and chord rotation $\psi$) and the loads. Then **joint equilibrium** equations give the unknown displacements.
- It is suitable for **continuous beams and rigid frames**, especially when $D_k < D_s$.

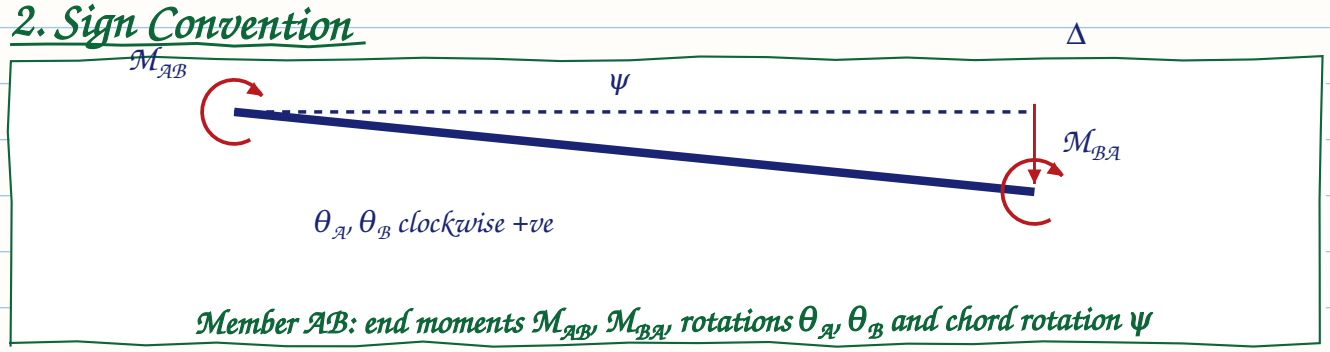
**Main idea**

Member end moment = (fixed end moment due to loads) + (moment due to end rotations and relative displacement).

### Assumptions

1. Material is linearly elastic; small deflections; $E$ and $I$ constant for each member.
2. Joints are **rigid** — all members meeting at a joint rotate through the same angle.
3. **Axial and shear deformations are neglected**; only bending deformation considered.
4. Distortion due to axial force is neglected, so the **length of members remains unchanged**.
5. Supports unyielding unless settlement is stated.

## 2. Sign Convention




**Member AB: end moments $M_{AB}$, $M_{BA}$, rotations $\theta_A$, $\theta_B$ and chord rotation $\psi$**

- **Clockwise** end moments (acting on the member), **clockwise** joint rotations $\theta_A$, $\theta_B$, and **clockwise chord rotation $\psi$** are all **positive**.
- $\psi = \Delta / L$, where $\Delta$ is the relative transverse displacement of the ends; it is **positive (clockwise)** when the far end **B moves downward** relative to A (or A rises relative to B).
- At a joint the moment of the member on the joint is opposite to the moment of the joint on the member (**Newton's third law**): $\Sigma M$ at a joint = 0.
- For BMD: a clockwise moment at the **left** end of a member or an anticlockwise moment at the

**right** end gives **sagging** .

## 3. Derivation of Slope-Deflection Equations

- Take a member AB of length $L$, constant $EI$, with end moments $M_{AB}$ and $M_{BA}$, end rotations $\theta_A$, $\theta_B$ and chord rotation $\psi$. Treat it as a simply supported beam of span $L$ carrying end moments ( **M/EI diagram = two triangles** ) — use the **moment-area / conjugate beam** method.
- The rotation of each end relative to the chord ($\theta - \psi$):

**From conjugate beam (end moments only)**

$$\theta_A - \psi = (L/6EI)(2M_{AB} - M_{BA})$$
$$\theta_B - \psi = (L/6EI)(2M_{BA} - M_{AB})$$

- Check: $M_{AB}$ alone $\rightarrow \theta_A = ML/3EI$ (clockwise) and the far end rotates $ML/6EI$ in the opposite sense.
- Solve for $M_{AB}$ and $M_{BA}$: multiply the first by 2 and add the second:
- $12EI(\theta_A - \psi)/L + 6EI(\theta_B - \psi)/L = 3M_{AB} \rightarrow$ **$M_{AB} = (2EI/L)(2\theta_A + \theta_B - 3\psi)$** ; similarly $M_{BA}$.

**Slope-deflection equations (with loads)**

$$M_{AB} = F_{AB} + (2EI/L)(2\theta_A + \theta_B - 3\psi)$$
$$M_{BA} = F_{BA} + (2EI/L)(2\theta_B + \theta_A - 3\psi)$$

- $F_{AB}$, $F_{BA}$ = **fixed end moments** due to the loads on the span (clockwise +ve). They are found by assuming both ends fixed (Unit III).
- Using **$k = I/L$** (relative stiffness): $M_{AB} = F_{AB} + 2EK(2\theta_A + \theta_B - 3\psi)$.

### Physical meaning of terms

| Term | Meaning |
|---|---|
| $4EI\theta_A/L$ | moment needed to rotate end A by $\theta_A$ with B fixed — **stiffness $4EI/L$** |
| $2EI\theta_B/L$ | **carry-over** moment: half of $4EI\theta/L$ goes to the far end (carry-over factor ½) |
| $-6EI\psi/L$ | moment due to relative displacement of the ends |
| $F_{AB}$ | fixed end moment due to loads |

## 4. Fixed End Moments (clockwise positive)

| Loading on span AB (length $L$) | $F_{AB}$ | $F_{BA}$ |
|---|---|---|
| UDL $w$ over whole span | $-wL^2/12$ | $+wL^2/12$ |
| Central point load $P$ | $-PL/8$ | $+PL/8$ |
| Point load $P$ at $a$ from A ($b$ from B) | $-Pab^2/L^2$ | $+Pa^2b/L^2$ |

| Loading on span AB (length L) | $F_{AB}$ | $F_{BA}$ |
|---|---|---|
| Two equal loads at L/3 and 2L/3 | $-2PL/9$ | $+2PL/9$ |
| UVL zero at A, w at B | $-wL^2/30$ | $+wL^2/20$ |
| UVL w at A, zero at B | $-wL^2/20$ | $+wL^2/30$ |
| Couple $M_0$ at a from A (clockwise) | $+M_0\,b(2a-b)/L^2$ | $+M_0\,a(2b-a)/L^2$ |
| Settlement Δ of B (downward), no load | $-6EI\Delta/L^2$ | $-6EI\Delta/L^2$ |

→ Remember: at the **left end the FEM of a downward load is anticlockwise (–)** and at the **right end clockwise (+)**. Equal magnitudes for symmetrical loads.

## 5. Modified Equation — Far End Hinged or Roller

- If the far end B of the member is **hinged or a roller** ($M_{BA} = 0$), substitute to remove $\theta_B$:

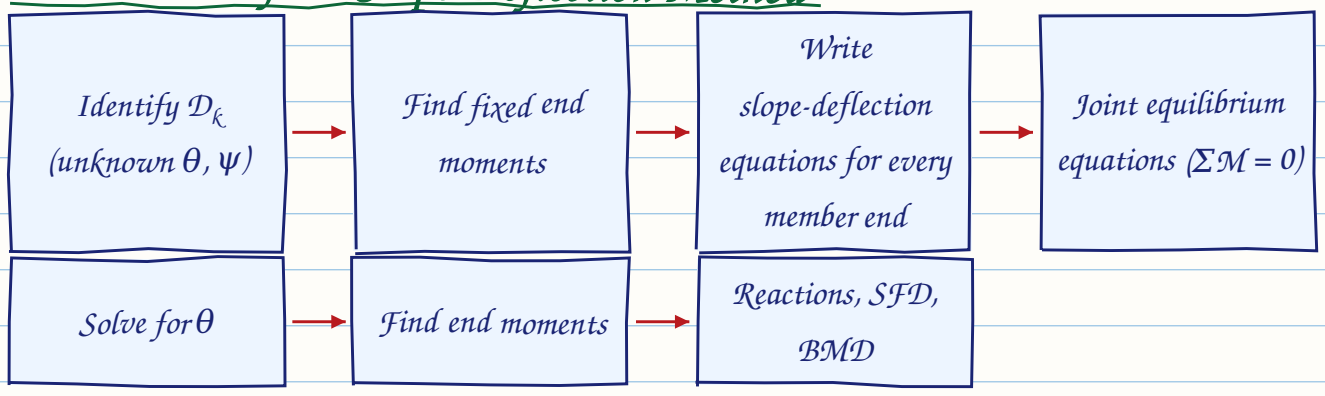
**Modified slope-deflection equation**

$$M_{AB} = F'_{AB} + (3EI/L)(\theta_A - \psi) \text{ where } F'_{AB} = F_{AB} - \tfrac{1}{2} F_{BA}$$

- Stiffness of the member becomes **3EI/L** (instead of 4EI/L); no carry-over to the hinged end.
- Modified fixed end moments: **UDL: $-wL^2/8$** ; central point load: **$-3PL/16$** .
- It reduces the number of unknowns — use it for the **end spans simply supported** .

| Far end condition | Stiffness | Carry-over |
|---|---|---|
| Fixed | 4EI/L | 1/2 |
| Hinged / roller | 3EI/L | 0 |
| Free (cantilever) | 0 (no stiffness) | — |

## 6. Procedure of the Slope-Deflection Method



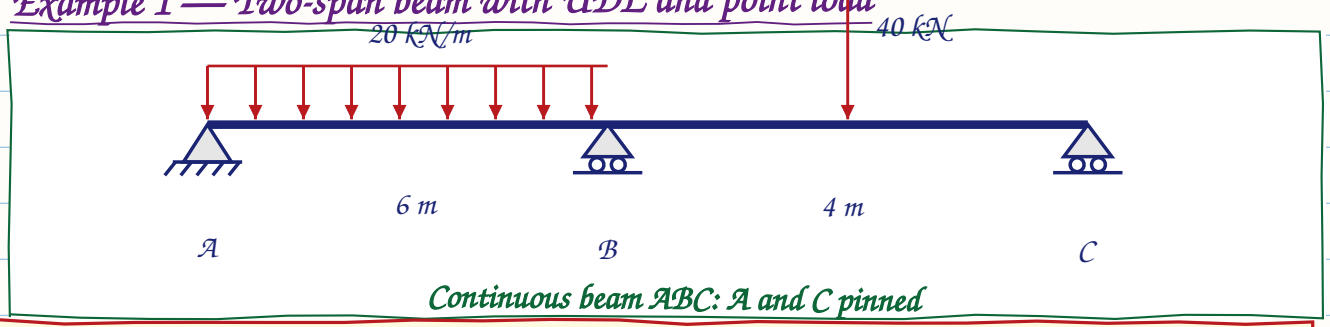

1. **Number of unknowns:** all joint rotations that are not zero (fixed ends have $\theta = 0$) and sway displacements.
2. Find $F_{AB}$, $F_{BA}$ for loaded spans, $\psi$ for members whose ends have relative displacement (settlement or sway).
3. Write the **two equations for each member** (or modified one if the far end is hinged).

4. Write **equilibrium equations** : at each joint $\Sigma M = 0$ (algebraic sum of the member end moments = 0, or = applied external moment). For a free-end overhang, the moment is known.
5. For frames with sway, add the **shear (storey) equation** ; in this unit sway is excluded.
6. Solve simultaneous equations ($EI\theta$ will come out); back-substitute for the end moments (check $\Sigma M$ at joints = 0).
7. Find **reactions** : $R$ = (free reaction) $\pm (M_{BA} + M_{AB})/L$; draw **SFD and BMD** .

→ Tip: Always work with $EI\theta$ as the unknown. Keep the units consistent (kN, m).

## 7. Continuous Beams without Settlement — Solved Examples

### Example 1 — Two-span beam with UDL and point load




Continuous beam ABC: A and C pinned

**Problem**

AB = 6 m carries UDL 20 kN/m; BC = 4 m carries a 40 kN load at its centre. EI is constant. Ends A and C are simply supported. Find the moments and reactions.

- Only unknown: $\theta_B$ (A and C hinged → use the modified equation).
- FEMs: $F_{BA} = +wL^2/12 = +60$, $F_{AB} = -60 \rightarrow F'_{BA} = 60 + 30 = +90$. For BC: $F_{BC} = -PL/8 = -20$, $F_{CB} = +20 \rightarrow F'_{BC} = -20 - 10 = -30$.
- **$M_{BA} = 90 + (3EI/6)\theta_B = 90 + 0.5\,EI\theta_B$ ; $M_{BC} = -30 + (3EI/4)\theta_B = -30 + 0.75\,EI\theta_B$**
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow 60 + 1.25\,EI\theta_B = 0 \rightarrow$ **$EI\theta_B = -48$** .
- $M_{BA} = 90 - 24 =$ **$+66$ kN·m** ; $M_{BC} = -30 - 36 =$ **$-66$ kN·m** . So **$M_B = 66$ kN·m (hogging)** ; $M_A = M_C = 0$.
- Reactions: $R_A = 60 - 66/6 =$ **49 kN** ; $R_B = (60 + 11) + (20 + 16.5) =$ **107.5 kN** ; $R_C = 20 - 66/4 =$ **3.5 kN** ; check: 49 + 107.5 + 3.5 = 160 = 120 + 40 √.
- Check by three-moment theorem: $2M_B(10) = 1080 + 240 \rightarrow M_B = 66$ √.
- Sagging in AB: $x = 49/20 = 2.45$ m from A → $M = 49 \times 2.45 - 10 \times 2.45^2 =$ **60.0 kN·m** . Under the 40 kN load (2 m from C): $M = 3.5 \times 2 =$ **7 kN·m** (small, since the hogging moment at B dominates).

### Example 2 — Beam with a fixed end

**Problem**

ABC with A fixed, B and C rollers, AB = BC = 6 m, UDL 20 kN/m on both spans.

- $\theta_A = 0$; unknown $\theta_B$ (C roller → modified for BC). $F_{AB} = -60$, $F_{BA} = +60$, $F'_{BC} = -60 - 30 =$

$-90$.

- $M_{AB} = -60 + (2EI/6)(\theta_B) = -60 + EI\theta_B/3$
- $M_{BA} = +60 + (2EI/6)(2\theta_B) = 60 + 2EI\theta_B/3$
- $M_{BC} = -90 + (3EI/6)\theta_B = -90 + 0.5\,EI\theta_B$
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow -30 + 1.1667\,EI\theta_B = 0 \rightarrow$ **$EI\theta_B = 25.71$** .

**Results**

**$M_{AB} = -51.43$, $M_{BA} = +77.14$, $M_{BC} = -77.14$ kN·m** (matches the three-moment result of Unit III)

- Span AB: $R_A = 60 + (51.43 - 77.14)/6 =$ **55.71 kN**, $R_B$(left) = 64.29 kN. Span BC: $R_C = 60 - 77.14/6 =$ **47.14 kN**, $R_B$(right) = 72.86 kN.
- **$R_B = 137.14$ kN**; check: 55.71 + 137.14 + 47.14 = 240 = total load √.

## Example 3 — Three-span continuous beam (equal spans, UDL)

- Spans $L$, UDL $w$, ends simply supported; symmetry: $\theta_B = -\theta_C$. Using modified equation for the end spans and full equation for the middle span:
- $M_{BA} = wL^2/8 + (3EI/L)\theta_B$ ; $M_{BC} = -wL^2/12 + (2EI/L)(2\theta_B + \theta_C) = -wL^2/12 + (2EI/L)\theta_B$.
- $\Sigma M_B = 0$: $wL^2/8 - wL^2/12 + (5EI/L)\theta_B = 0 \rightarrow EI\theta_B = -wL^3/120$.
- **$M_{BA} = wL^2/8 - wL^2/40 = wL^2/10$** √ (matches three-moment result).

## Example 4 — Beam with overhang

- A cantilever overhang CD: its end moment at C is fixed: **$M_{CD} = -$ (load × arm)** ; write only $M_{CB}$ equation with this known moment.
- Joint C equation: $M_{CB} + M_{CD} = 0 \rightarrow M_{CB} = +$ load × arm.

# 8. Continuous Beams with Settlement of Supports

- A support settlement $\Delta$ produces a **chord rotation** $\psi = \Delta / L$ in the adjacent spans.

| Case | $\psi$ |
|---|---|
| Span AB: B lower than A by $\Delta$ | $\psi_{AB} = +\Delta / L$ (clockwise) |
| Span BC: C higher than B (B lower) | $\psi_{BC} = -\Delta / L$ (anticlockwise) |
| Both ends settle equally | $\psi = 0$ |

- $\psi$ is **the same for both ends** of a member.

## Example 5 — Settlement of the middle support

**Problem**

Two equal spans $AB = BC = 6$ m, UDL 20 kN/m, $EI = 30000$ kN·m². A and C pinned. Support B settles by 10 mm. Find the support moment.

- $\psi_{AB} = +0.01/6 = 0.001667$ ; $\psi_{BC} = -0.001667$.
- $M_{BA} = F'_{BA} + (3EI/L)(\theta_B - \psi_{AB}) = 90 + 15000\,(\theta_B - 0.001667)$
- $M_{BC} = F'_{BC} + (3EI/L)(\theta_B - \psi_{BC}) = -90 + 15000\,(\theta_B + 0.001667)$
- Joint B: $30000\,\theta_B = 0 \rightarrow \theta_B = 0 \rightarrow M_{BA} = 90 - 25 =$ **65 kN·m** ; $M_{BC} = -90 + 25 =$ **– 65 kN·m** .
- Without settlement 90 kN·m; **settlement reduces the support moment** by $3EI\Delta/L^2 = 25$ kN·m.
- $R_A = 60 - 65/6 =$ **49.17 kN**, $R_C =$ **49.17 kN**, $R_B = 240 - 98.33 =$ **141.67 kN**.

## Example 6 — Settlement of one end support

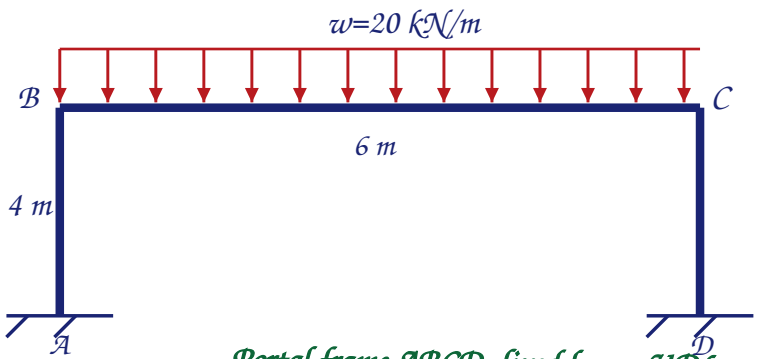
- If the end support C settles by Δ (downward), only span BC gets a chord rotation: C is lower than B, so the chord rotates clockwise and **$\psi_{BC} = +\Delta / L$** . Solve with $\theta_B$ as the unknown in the same way.

→ Always decide the sign of ψ with a quick sketch: **the member rotates clockwise** → ψ positive.

# 9. Single-Bay Portal Frames without Sway

- A portal frame is **without sway** if the frame and its loading are **symmetrical** or if the lateral movement of the beam is **prevented** by a support.
- Then the chord rotation $\psi = 0$ for **all** members (columns do not sway, beam has no differential movement).
- **Joint rotations $\theta_B$ and $\theta_C$ are the unknowns** (fixed bases: $\theta_A = \theta_D = 0$); for hinged bases use the modified equation for the columns.
- For symmetrical loading: **$\theta_C = -\theta_B$** — only one unknown.

## Example 7 — Fixed-base portal frame with UDL on the beam




Portal frame ABCD, fixed bases, UDL on beam BC

**Problem**

Columns AB and DC are 4 m high; beam BC = 6 m; UDL 20 kN/m on BC; EI is the same for all members. A and D fixed.

- Symmetry → $\theta_C = -\theta_B$. FEMs for BC: $F_{BC} = -60$, $F_{CB} = +60$.
- $M_{AB} = (2EI/4)(\theta_B) = 0.5\,EI\theta_B$ ; $M_{BA} = (2EI/4)(2\theta_B) = EI\theta_B$
- $M_{BC} = -60 + (2EI/6)(2\theta_B + \theta_C) = -60 + (EI/3)\,\theta_B$
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow EI\theta_B\,(1 + 1/3) = 60 \rightarrow$ **$EI\theta_B = 45$** .

**End moments**

**$M_{AB} = 22.5$ kN·m, $M_{BA} = 45$, $M_{BC} = -45$, $M_{CB} = +45$, $M_{CD} = -45$, $M_{DC} = -22.5$ kN·m**

- Horizontal reactions: $H_A = H_D = (M_{AB} + M_{BA})/h = 67.5/4 =$ **16.88 kN** (acting inwards at the bases). Vertical: $V_A = V_D = wL/2 =$ **60 kN**.
- BMD: beam hogging **45 kN·m** at each end; centre sagging = $wL^2/8 - 45 = 90 - 45 =$ **+ 45 kN·m** ; columns: 22.5 at the base and 45 at the top.
- Check: $H_A + H_D$ horizontal forces balance (equal and opposite); $\Sigma V = 120$ = total load √.

## Example 8 — Beam stiffer than columns ($I_{beam} = 2\,I_{column}$)

- $M_{BA} = EI\theta_B$ (columns, EI/4 stiffness); $M_{BC} = -60 + (2 \times 2EI/6)(2\theta_B + \theta_C) = -60 + (2EI/3)\,\theta_B$.
- Joint B: $EI\theta_B\,(1 + 2/3) = 60 \rightarrow EI\theta_B = 36 \rightarrow$ **$M_{BA} = 36$, $M_{AB} = 18$, $M_{BC} = -36$ kN·m** .
- **Stiffer beam → larger share of the load moment attracted by the beam; column moments smaller** (compare 45 earlier).

## Example 9 — Hinged-base portal frame (symmetric UDL)

- Columns hinged at A and D → modified: $M_{BA} = (3EI/4)\theta_B$ ($F'=0$). $M_{BC} = -60 + (EI/3)\theta_B$.
- Joint B: $EI\theta_B\,(0.75 + 0.333) = 60 \rightarrow EI\theta_B = 55.38 \rightarrow$ **$M_{BA} = 41.54$ kN·m** , $M_{BC} = -41.54$ kN·m; $M_{AB} = 0$.
- Horizontal thrust $H = 41.54/4 =$ **10.38 kN**; beam centre moment = 90 − 41.54 = **48.46 kN·m** . Compared with the fixed-base frame: smaller thrust and column moment, but **larger sagging moment in the beam** .

# 9A. More Solved Examples — Beams

## Example 10 — Propped cantilever (A fixed, B roller) with UDL

- $\theta_A = 0$, $M_{BA} = 0 \rightarrow$ use the modified equation for end A: $M_{AB} = F_{AB} - \tfrac{1}{2}F_{BA} + (3EI/L)(\theta_A - \psi)$.
- $M_{AB} = -wL^2/12 - \tfrac{1}{2}\,(wL^2/12) + 0 =$ **$-wL^2/8$** (hogging at the fixed end) √ — result in one step.
- $R_A = 5wL/8$, $R_B = 3wL/8$.

### Example 11 — Two spans with different spans and different EI

**Problem**

$AB = 6$ m ($EI$), $BC = 4$ m ($2EI$); UDL 20 kN/m on both spans; $A$ and $C$ simply supported. Find $M_B$.

- $F'_{BA} = wL^2/8 = 20 \times 36/8 = 90$ ; $F'_{BC} = -20 \times 16/8 = -40$.
- $M_{BA} = 90 + (3EI/6)\theta_B = 90 + 0.5\,EI\theta_B$
- $M_{BC} = -40 + (3 \times 2EI/4)\theta_B = -40 + 1.5\,EI\theta_B$
- Joint $B$: $50 + 2\,EI\theta_B = 0 \rightarrow EI\theta_B = -25$.

**Result**

**$M_{BA} = 90 - 12.5 = 77.5$ kN·m ; $M_{BC} = -40 - 37.5 = -77.5$ kN·m** → $M_B = 77.5$ kN·m hogging.

- $R_A = 60 - 77.5/6 =$ **47.08 kN**; $R_C = 40 - 77.5/4 =$ **20.63 kN**; $R_B = 200 - 67.71 =$ **132.29 kN** .
- Observation: the **stiffer span attracts more moment** , as stiffness $= 3EI/L$ is larger for $BC$.

### Example 12 — Three equal spans with central point loads

- Spans $L$, central load $P$ in each span, ends pinned; $\theta_C = -\theta_B$ (symmetry).
- $F'_{BA} = PL/8 + \tfrac{1}{2}(PL/8) = 3PL/16$ ; $M_{BA} = 3PL/16 + (3EI/L)\theta_B$.
- $M_{BC} = -PL/8 + (2EI/L)(2\theta_B + \theta_C) = -PL/8 + (2EI/L)\theta_B$.
- Joint $B$: $PL/16 + 5EI\theta_B/L = 0 \rightarrow EI\theta_B = -PL^2/80$.

**Result**

**$M_B = M_C = 3PL/20$** — for $L = 6$ m and $P = 40$ kN: **36 kN·m (hogging)** ; $R_A = P/2 - M_B/L = 20 - 6 =$ **14 kN** $= R_D$; $R_B = R_C = (3P - 28)/2 =$ **46 kN** (check: $14 + 46 + 46 + 14 = 120 = 3P$ √).

### Example 13 — Joint carrying an external moment

- If a couple $M_0$ (clockwise) acts at joint $B$, the equilibrium equation becomes **$M_{BA} + M_{BC} = M_0$** (the joint moment is balanced by the member moments).

## 9B. Portal Frame — Extra Example and BMD

### Example 14 — Fixed-base portal with a central point load on the beam

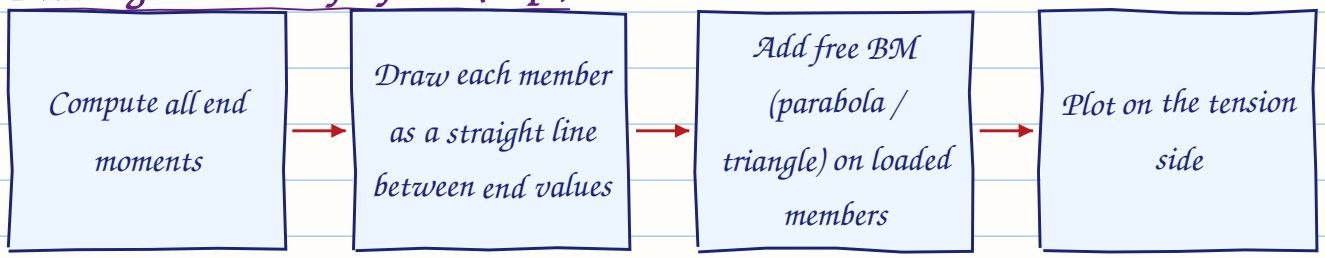
**Problem**

$h = 4$ m, $L = 6$ m, $EI$ same; $P = 48$ kN at the centre of $BC$. $A$ and $D$ fixed.

- $F_{BC} = -PL/8 = -36$. By symmetry $\theta_C = -\theta_B$.
- $M_{BA} = EI\theta_B$ ; $M_{BC} = -36 + (EI/3)\theta_B$. Joint $B$: $(4/3)\,EI\theta_B = 36 \rightarrow$ **$EI\theta_B = 27$** .
- **$M_{BA} = 27$, $M_{AB} = 13.5$, $M_{BC} = -27$ kN·m** . Beam centre: $PL/4 - 27 = 72 - 27 =$ **+ 45 kN·m** .

- Horizontal reaction $H = (13.5 + 27)/4 =$ **10.13 kN**; $V_A = V_D =$ **24 kN**.

### Drawing the BMD of a frame (steps)




- Plot the BM on the **tension side** of each member; at a rigid joint the BM is **continuous** in value — the moment going into the joint from one member equals the moment leaving by the other (joint equilibrium).
- Check: sum of moments at each joint = 0 and the horizontal reactions at the bases are equal and opposite (no sway).

| Frame result | Fixed bases | Hinged bases | Remark |
|---|---|---|---|
| Column moment at base | 22.5 | 0 | hinged base has no moment |
| Moment at joint B | 45 | 41.5 | for UDL 20 kN/m, $h = 4$, $L = 6$ |
| Beam centre BM | 45 | 48.5 | beam works harder when bases are hinged |
| Horizontal reaction | 16.9 | 10.4 | thrust at bases |

## 9C. Fixed Beams and Propped Cantilever by Slope-Deflection (Quick Methods)

- **Fixed beam:** $\theta_A = \theta_B = 0$ and no settlement → end moments = fixed end moments $F_{AB}$, $F_{BA}$ directly.
- **Fixed beam with settlement Δ:** $M_{AB} = F_{AB} - 6EI\Delta/L^2$, $M_{BA} = F_{BA} - 6EI\Delta/L^2$ (B sinks).
- **Support rotation $\theta_A$:** $M_{AB} = F_{AB} + 4EI\theta_A/L$, $M_{BA} = F_{BA} + 2EI\theta_A/L$.
- **Propped cantilever with point load P at the centre:** $M_{AB} = F_{AB} - \frac{1}{2}F_{BA} = -PL/8 - PL/16 =$ **$-3PL/16$** √.

## 9D. Checks and Common Mistakes

| No. | Point to check |
|---|---|
| 1 | Clockwise = positive for moments, θ and ψ — be consistent throughout. |
| 2 | FEM of a downward load: **− at the left end, + at the right end** of the span. |
| 3 | Use the **modified** equation only when the far end is hinged/roller and write $F - \frac{1}{2}F'$ correctly. |

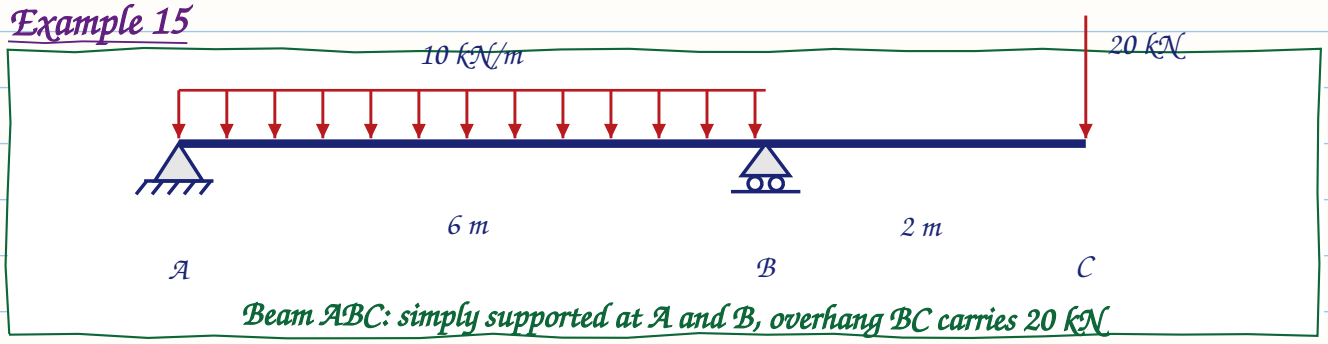
| No. | Point to check |
|---|---|
| 4 | $\psi$ is the same for both ends of the member; sign by a sketch. |
| 5 | Joint equation: $\Sigma M$ (member end moments at the joint) = 0, not the sum of rotations. |
| 6 | The moment at an overhang end is known — do not treat it as unknown. |
| 7 | Verify by a second method (three-moment equation or $\Sigma V$ = total load). |
| 8 | Cantilever overhang members have **zero stiffness** ; do not include them in $EI\theta$ terms. |

## 9E. Practice Problems with Answers

| No. | Problem | Answer |
|---|---|---|
| 1 | Two equal spans 5 m, UDL 24 kN/m, A and C pinned | $M_B = wL^2/8 = 75$ kN·m |
| 2 | Propped cantilever $L = 4$ m, UDL 15 kN/m (fixed at A) | $M_A = wL^2/8 = 30$ kN·m |
| 3 | Three equal spans 6 m, UDL 10 kN/m, ends pinned | $M_B = M_C = wL^2/10 = 36$ kN·m |
| 4 | Two equal spans 6 m, central load 60 kN in each | $M_B = 3PL/16 = 67.5$ kN·m |
| 5 | Fixed-base portal $h = 4$ m, $L = 6$ m, UDL 30 kN/m (EI same) | $M_{BA} = 67.5$; $M_{AB} = 33.75$ kN·m |
| 6 | Two equal spans 6 m, UDL 20 kN/m, B settles 10 mm, $EI = 30000$ | $M_B = 65$ kN·m |

## 9F. Beam with Overhang — Solved Example

### Example 15




**Beam ABC: simply supported at A and B, overhang BC carries 20 kN**

- Overhang moment: $M_{BC} = -20 \times 2 = -40$ kN·m (known; the overhang is a cantilever, no stiffness).
- Joint B: $M_{BA} + M_{BC} = 0 \rightarrow M_{BA} = +40$ kN·m .
- Check the unknown rotation: $M_{BA} = F'_{BA} + (3EI/6)\theta_B = 45 + 0.5\,EI\theta_B = 40 \rightarrow EI\theta_B = -10$ (anticlockwise).
- Reactions: $R_A = 30 - 40/6 =$ **23.33 kN**; $R_B = (30 + 6.67) + 20 =$ **56.67 kN**; check: $23.33 + 56.67 = 80 = 60 + 20$ √.
- Max sagging in AB at $x = 23.33/10 = 2.33$ m: $M = 23.33 \times 2.33 - 5 \times 2.33^2 =$ **27.2 kN·m** .

## 9G. Shear Force and Bending Moment Values for Example 1

| Section | Shear force (kN) | Bending moment (kN·m) |
|---|---|---|
| A | + 49 | 0 |
| 2.45 m from A | 0 | + 60.0 (max sagging) |
| B (left) | − 71 | − 66 |
| B (right) | + 36.5 | − 66 |
| under 40 kN load (2 m from B) | + 36.5 → − 3.5 | + 7.0 |
| C | − 3.5 | 0 |

- Check: area of SFD between B and the point of zero moment gives BM change; BM is maximum where SF changes sign.

## 9H. Stiffness and Carry-Over — Quick Derivation

- Apply a unit rotation $\theta_A = 1$ at A while B is **fixed** ($\theta_B = 0$, $\psi = 0$): $M_{AB} = 4EI/L$ (stiffness) and $M_{BA} = 2EI/L \rightarrow$ **carry-over factor = ½** .
- If B is **hinged** : $M_{BA} = 0 \rightarrow$ from the two equations $2\theta_B + \theta_A = 0 \rightarrow \theta_B = -\theta_A/2 \rightarrow M_{AB} = (2EI/L)(2\theta_A - \theta_A/2) = 3EI\,\theta_A / L$ .
- Relative displacement: $M_{AB} = M_{BA} = -6EI\psi/L \rightarrow$ settlement $\Delta$ gives $6EI\Delta/L^2$.
- These three results ($4EI/L$, $3EI/L$, $6EI/L^2$) are the entries of the **member stiffness matrix** — the slope-deflection method is the hand form of the **matrix stiffness method** .

| Quantity | Far end fixed | Far end hinged |
|---|---|---|
| Rotation stiffness at near end | $4EI/L$ | $3EI/L$ |
| Carry-over moment | ½ M | 0 |
| Moment for settlement $\Delta$ (per unit $\Delta$) | $-6EI/L^2$ (both ends) | $-3EI/L^2$ (near end only) |

## 9I. Choosing a Method — Quick Guide

Count $D_s$ and $D_k$ → Few redundants ($D_s$ small)? Use force methods → Few joint displacements ($D_k$ small)? Use slope-deflection / moment distribution

- **Fixed beam:** $D_s = 3$, $D_k = 0 \rightarrow$ slope-deflection is immediate (no unknowns).
- **Two-span beam with simple ends:** $D_k = 1$ ($\theta_B$) → one equation.
- **Portal frame without sway (symmetric):** $D_k = 1$ effectively.

## 10. Frames with Unsymmetrical Loading (Notes)

- If the frame is unsymmetrical in loading but sway is prevented, **$\theta_B$ and $\theta_C$ are both unknown**; write two joint equations, solve simultaneously.
- If the frame **sways**, an extra unknown $\Delta$ ($\psi$ for columns) appears, and an additional equation is needed — the **horizontal force (storey shear) equation** $\Sigma H = 0$. This is outside the present scope.
- Typical example: a portal with a horizontal load at beam level sways → solve with sway.

## 11. Relative Stiffness and Distribution (link to Unit V)

- Joint equilibrium shows that a joint moment $M$ is shared in proportion to **stiffness $4EI/L$ (or $3EI/L$)**: this idea leads to **moment distribution** (Unit V) — a stepwise (iterative) form of slope deflection.
- Distribution factor $DF$ = stiffness of member ÷ sum of stiffnesses at the joint.

## 12. Advantages and Limitations

| Aspect | Advantages | Limitations |
|---|---|---|
| Method | systematic; works for beams and frames | simultaneous equations if many joints |
| Unknowns | only joint displacements ($D_k$) | large number for tall frames |
| Settlement | easy to include ($\psi$) | sign errors are common |
| Computer use | forms the basis of the matrix stiffness method | hand work lengthy for sway frames |

## 13. Slope-Deflection vs Moment Distribution vs Three-Moment

| Basis | Slope-deflection | Three-moment | Moment distribution |
|---|---|---|---|
| Unknowns | $\theta$, $\psi$ | support moments | none (iterative) |
| Equations | joint equilibrium | one per pair of spans | none |
| Best for | beams and frames | continuous beams | beams and frames |
| Result | exact | exact | exact (after convergence) |

## 14. Quick Revision — Unit IV

**Formulae**

$M_{AB} = F_{AB} + (2EI/L)(2\theta_A + \theta_B - 3\psi)$ | $M_{BA} = F_{BA} + (2EI/L)(2\theta_B + \theta_A - 3\psi)$ | Far end hinged: $M_{AB} = F_{AB} - \tfrac{1}{2}F_{BA} + (3EI/L)(\theta_A - \psi)$ | $\psi = \Delta/L$ (clockwise +ve) | FEM: $\mp wL^2/12$, $\mp PL/8$, $\mp Pab^2/L^2$, $\pm Pa^2b/L^2$ | Joint: $\Sigma M = 0$.

## Possible exam questions

1. Derive the slope-deflection equations for a member with end moments.
2. Analyse a two-span continuous beam by the slope-deflection method; draw BMD.
3. Analyse the continuous beam when one support settles.
4. Analyse a single-bay portal frame without sway using the slope-deflection method.
5. Explain the modified slope-deflection equation for a hinged far end.

# UNIT - V

## Moment Distribution Method

## 1. Introduction

- The **moment distribution method (MDM)** was developed by **Prof. Hardy Cross** in **1930** (ASCE Transactions, 1932). It is an **iterative displacement method** — a convenient hand-calculation form of the slope-deflection method.
- It avoids solving simultaneous equations: the end moments are obtained by a series of **successive approximations** (cycles) that converge quickly.
- Used for **continuous beams and rigid frames** ; popular in design offices before computers; still used for quick checks.

**Basic idea**

Lock all joints against rotation → calculate fixed end moments → release one joint at a time, **distribute the unbalanced moment** to the members meeting at the joint, **carry over** half to the far ends → repeat until the unbalanced moments become negligible.

### Assumptions

- Prismatic members (constant $EI$), linear elastic behaviour, small deflections.
- Axial and shear deformations are neglected; joints are rigid.
- Applicable directly to structures **without sway** ; sway frames need an extra correction.

## 2. Basic Definitions

| Term | Meaning |
|---|---|
| Fixed end moment (FEM) | moment developed at the ends of a loaded member when both its ends are fixed (Unit III table) |
| Unbalanced moment | algebraic sum of the FEMs (or current end moments) of the members meeting at a joint |
| Stiffness ($K$) | moment required to produce a **unit rotation** at one end, the far end being fixed ($4EI/L$) or hinged ($3EI/L$) |
| Relative stiffness | ratio of stiffnesses; $I/L$ can be used when $E$ is constant |
| Distribution factor (DF) | fraction of the unbalanced moment taken by a member: $DF = K_{member} / \Sigma K$ at the joint |
| Distributed (balancing) moment | − (unbalanced moment) × DF |

| Term | Meaning |
|---|---|
| Carry-over factor (COF) | ratio of the moment induced at the far fixed end to the moment applied at the near end = ½ |
| Carry-over moment | COF × distributed moment, placed at the far end |

### Stiffness and carry-over (from slope-deflection)

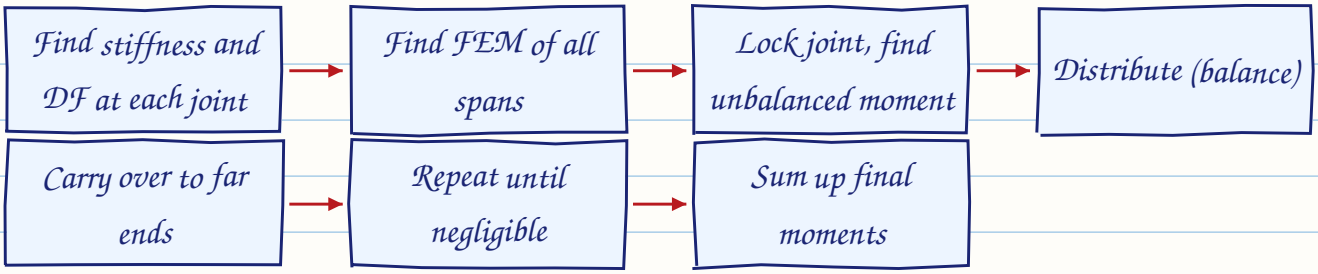
| Far end | Stiffness K | Carry-over factor |
|---|---|---|
| Fixed | $4EI / L$ | + 1/2 |
| Hinged / roller (simply supported) | $3EI / L$ (= ¾ × $4EI/L$) | 0 |
| Free end (cantilever) | 0 | — |
| Symmetric loading, symmetric span | $2EI / L$ | — |
| Antisymmetric loading, symmetric span | $6EI / L$ | — |

- **Distribution factors at a joint always add up to 1.0** (check!).
- **DF of a fixed support = 0** (infinite stiffness — it does not rotate); **DF of a hinged/roller end support = 1.0** .
- Overhanging cantilever: moment at the support is **known** ; it enters as an unbalanced moment and has no DF.

## 3. Sign Convention

- **Clockwise end moments (acting on the member ends)** are **positive** ; anticlockwise negative (same as slope deflection).
- Moment acting **on the joint** from a member is opposite in sense to the moment acting on the member end.
- At a joint the sum of the member end moments must be **zero** at the end of the process (equilibrium).
- Balancing moment has the **opposite sign** of the unbalanced moment.
- For the final BMD: end moments of a member in the usual convention — **hogging** at an interior support for gravity loads.

## 4. Procedure of Moment Distribution




1. **Stiffness and DF:** compute $K$ for every member at each joint ($3EI/L$ if far end hinged; $4EI/L$ otherwise); find $DF = K/\Sigma K$.
2. **FEM:** for all loaded spans (clockwise +ve). For a settlement, add $FEM = \pm 6EI\Delta/L^2$.
3. Set up a **table** with a column for each member end (AB, BA, BC, CB ...).
4. **Balance** each joint in turn: unbalanced moment = $\Sigma$ moments at the joint; balancing moment = − unbalanced × DF; write under each member and draw a line.
5. **Carry over** half of each balancing moment to the other end of the member (not for hinged far end).
6. Continue until the **carry-over is negligible** (say less than 1% of the FEM).
7. **Add each column** to get final end moments; check $\Sigma M$ at joints = 0.
8. Compute **reactions** ($R$ = free reaction $\pm (M_{left} - M_{right})/L$), SFD and BMD.

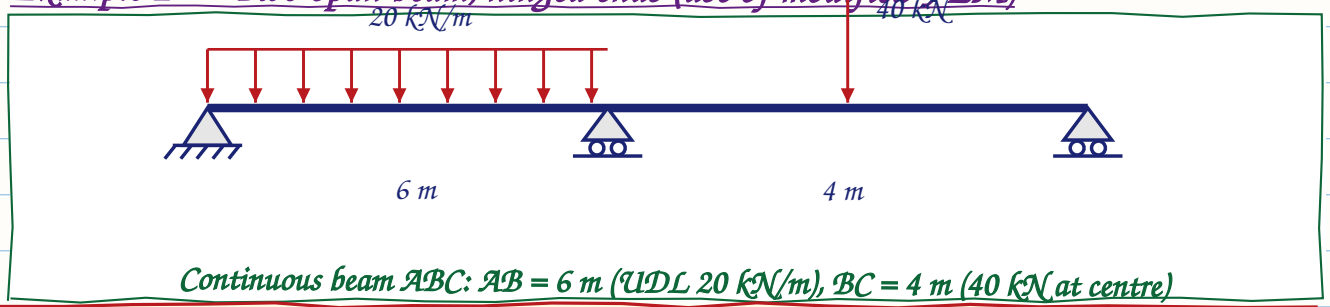
→ Shortcut: when the far end is **hinged**, use the **modified FEM** $F' = F - \frac{1}{2}F_{far}$ and stiffness $3EI/L$, and **do not carry over** to that end. A fixed end is never balanced ($DF = 0$); an **end hinged support** can be balanced once at the start.

### Order of balancing

- Any order is allowed, but balance the joint with the **largest unbalanced moment** first; it converges faster. For a joint where the unbalanced moment is zero skip it.

## 5. Application to Continuous Beams — Examples

### Example 1 — Two-span beam, hinged ends (use of modified FEM)




Continuous beam ABC: AB = 6 m (UDL 20 kN/m), BC = 4 m (40 kN at centre)

**Problem**

Same beam as Unit IV (EI constant). A and C simply supported.

- Both far ends hinged: $K_{BA} = 3EI/6 = 0.5\ EI$ ; $K_{BC} = 3EI/4 = 0.75\ EI$. **$DF_{BA} = 0.5/1.25 = 0.4$; $DF_{BC} = 0.6$.**
- FEM: AB = − 60, BA = + 60, BC = − 20, CB = + 20. Modified: $F'_{BA} = 60 + 30 =$ **+ 90** , $F'_{BC}$

= − 20 − 10 = **− 30** (no carry-over to A or C).

| Joint / member | BA | BC |
|---|---|---|
| DF | 0.4 | 0.6 |
| Modified FEM | + 90 | − 30 |
| Unbalanced at B | + 60 | |
| Balancing moment | − 24 | − 36 |
| Final moment | **+ 66** | **− 66** |

- **$M_B$ = 66 kN·m (hogging)** √ same as the slope-deflection and three-moment results. Reactions: $R_A$ = 49 kN, $R_B$ = 107.5 kN, $R_C$ = 3.5 kN.

## Example 2 — Beam with a fixed end (single cycle)

**Problem**

ABC; A fixed, B and C rollers; AB = BC = 6 m; UDL 20 kN/m on both spans.

- $K_{BA} = 4EI/6 = 0.667\,EI$, $K_{BC} = 3EI/6 = 0.5\,EI$ (C hinged). **$DF_{BA}$ = 0.571, $DF_{BC}$ = 0.429.**
- FEM: AB = − 60, BA = + 60, BC (modified) = − 60 − 30 = − 90.

| Member end | AB | BA | BC |
|---|---|---|---|
| DF | — | 0.571 | 0.429 |
| FEM | − 60 | + 60 | − 90 |
| Balance B (unbalanced − 30) | | + 17.14 | + 12.86 |
| Carry-over | + 8.57 | | |
| Final | **− 51.43** | **+ 77.14** | **− 77.14** |

- √ Equal to the slope-deflection result (Unit IV, Example 2).

## Example 3 — Three spans, two joints to balance (several cycles)

**Problem**

Beam ABCD; A fixed; B, C and D rollers; AB = BC = CD = 6 m; UDL 20 kN/m on all spans.

- D is hinged → modified stiffness for CD (3EI/6) and modified FEM $F'_{CD}$ = − 60 − 30 = − 90. $K_{BA} = K_{BC} = 4EI/6$ → **$DF_B$ = 0.5, 0.5** ; $K_{CB} = 4EI/6$, $K_{CD} = 3EI/6$ → **$DF_C$ = 0.571 (CB), 0.429 (CD)** .
- Joint B is balanced at the start (+ 60 − 60 = 0), so begin with C.

| Step | AB | BA | BC | CB | CD |
|---|---|---|---|---|---|
| DF | — | 0.500 | 0.500 | 0.571 | 0.429 |
| FEM | -60.00 | +60.00 | -60.00 | +60.00 | -90.00 |
| Bal C(1) | | | | +17.14 | +12.86 |
| CO(1) | | | +8.57 | | |

| Step | AB | BA | BC | CB | CD |
|---|---|---|---|---|---|
| Bal B(2) | | -4.29 | -4.29 | | |
| CO(2) | -2.14 | | | -2.14 | |
| Bal C(2) | | | | +1.22 | +0.92 |
| CO(2c) | | | +0.61 | | |
| Bal B(3) | | -0.31 | -0.31 | | |
| CO(3) | -0.15 | | | -0.15 | |
| Bal C(3) | | | | +0.09 | +0.07 |
| CO(3c) | | | +0.04 | | |
| Final (≈) | -62.30 | +55.41 | -55.36 | +76.16 | -76.16 |
| Exact | -62.31 | +55.38 | -55.38 | +76.15 | -76.15 |

- **Final:** $M_{AB} = -62.31$, $M_{BA} = 55.38$, $M_{BC} = -55.38$, $M_{CB} = 76.15$, $M_{CD} = -76.15$ kN·m — convergence is rapid (three cycles are enough).
- Check: $\Sigma M$ at $B = 55.38 - 55.38 = 0$ √; at $C = 76.15 - 76.15 = 0$ √.

### Example 4 — Beam with an overhang

**Problem**

$AB = 6$ m simply supported at A and B with UDL 10 kN/m; overhang $BC = 2$ m with 20 kN at C.

- Overhang moment at B: $M_{BC} = -20 \times 2 = -40$ (known); stiffness of the cantilever = 0, so $DF_{BA} = 1.0$.
- $F'_{BA} = wL^2/8 = +45$. Unbalanced at $B = 45 - 40 = +5 \rightarrow$ balancing $= -5 \rightarrow M_{BA} = +40$, $M_{BC} = -40$ (hogging 40 kN·m).
- $R_A = 30 - 40/6 =$ **23.33 kN**; $R_B = 36.67 + 20 =$ **56.67 kN** √ (same as Unit IV, Example 15).

## 6. Continuous Beams with Settlement of Supports

- A settlement $\Delta$ of a support introduces additional FEMs in the adjacent spans (chord rotation): $FEM = \pm 6EI\Delta/L^2$ for both ends of a span (same sign at both ends).
- **Sign:** when the right end B is **lower** than A, the chord rotates **clockwise** : $FEM_{AB} = FEM_{BA} = -6EI\Delta/L^2$ (anticlockwise on the member ends). When B is **higher** : $+6EI\Delta/L^2$.
- Add the settlement FEM to the FEM due to loads, then distribute as usual.
- If the settlement is given as a **ratio of EI** (e.g., $\Delta = 10$ mm, $EI = 30000$ kN·m²), compute $6EI\Delta/L^2$ numerically first.

### Example 5 — Middle support settles

**Problem**

Two equal spans AB = BC = 6 m, UDL 20 kN/m, EI = 30000 kN·m², A and C hinged; B settles by 10 mm.

- $6EI\Delta/L^2 = 6 \times 30000 \times 0.01 / 36 =$ **50 kN·m** .
- Span AB (B lower): $FEM_{AB} = -60 - 50 = -110$ ; $FEM_{BA} = +60 - 50 = +10$.
- Span BC (C higher than B): $FEM_{BC} = -60 + 50 = -10$ ; $FEM_{CB} = +60 + 50 = +110$.
- Modified (A, C hinged): $F'_{BA} = 10 + \frac{1}{2}(110) =$ **+ 65** ; $F'_{BC} = -10 - \frac{1}{2}(110) =$ **− 65** .
- $DF_{BA} = DF_{BC} = 0.5$. Unbalanced moment at B = 65 − 65 = 0 → **$M_{BA} = +65$, $M_{BC} = -65$ kN·m** .
- Compare: without settlement $M_B = 90$ kN·m; **settlement reduces the support moment by 25 kN·m** √ (same as Unit III and Unit IV).
- $R_A = 60 - 65/6 = 49.17$ kN; $R_B = 141.67$ kN.

## Example 6 — Settlement of an end support (outline)

- Fixed beam A–B with support B settled by Δ: $FEM_{AB} = FEM_{BA} = -6EI\Delta/L^2$ — since both ends are fixed no distribution takes place; these are the final moments (Unit III).
- For a continuous beam, if only C (end support) sinks, only the span BC gets the extra FEMs; B is then balanced with a non-zero unbalanced moment.

## 7. Symmetry in Moment Distribution

- For a **symmetrical structure with symmetrical loading** , the rotations at corresponding joints are equal and opposite. Use half the structure with modified stiffness:

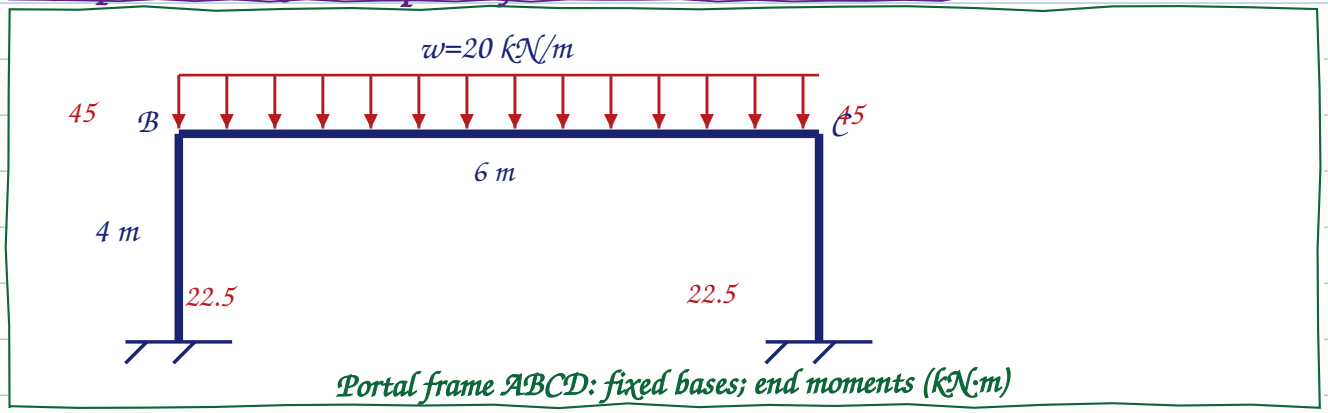
| Condition | Stiffness of the symmetric span (far end rotates in opposite direction) | Carry-over |
|---|---|---|
| Symmetric loading | 2EI/L | none to the far end (use mirror image) |
| Antisymmetric loading | 6EI/L | none to the far end |

- This reduces the number of joints to be balanced and often yields the answer in **one cycle** .

## 8. Single-Bay Portal Frames without Sway

- Frame is **without sway** if (i) the loading and the frame are symmetrical about the vertical axis, or (ii) lateral movement is prevented by a support.
- All members have **no chord rotation** ; fixed end moments act only on loaded members (columns usually have none).
- Steps: DF at B and C, FEM of the beam, balance joints, carry over to the column bases, repeat.

## Example 7 — Fixed-base portal frame with UDL on the beam




Portal frame ABCD: fixed bases; end moments (kN·m)

- $h = 4$ m, $L = 6$ m, $EI$ constant. Use symmetry: **beam stiffness $= 2EI/L = 0.333\,EI$ ; column stiffness (base fixed) $= 4EI/h = EI$ .**
- **$DF_{BA} = 1/(1 + 0.333) = 0.75$; $DF_{BC} = 0.25$.**
- $FEM_{BC} = -wL^2/12 = -60$ (and $FEM_{CB} = +60$ by symmetry).

| Member end | AB | BA | BC |
|---|---|---|---|
| DF | — | 0.75 | 0.25 |
| FEM | 0 | 0 | − 60 |
| Balance at B (+ 60) | | + 45 | + 15 |
| Carry-over (to base A) | + 22.5 | | |
| Final | **+ 22.5** | **+ 45** | **− 45** |

- By symmetry: $M_{CB} = +45$, $M_{CD} = -45$, $M_{DC} = -22.5$ kN·m.
- Beam centre $BM = wL^2/8 - 45 =$ **+ 45 kN·m** ; $H_A = H_D = (22.5 + 45)/4 =$ **16.88 kN**; $V = 60$ kN.
- √ Same as the slope-deflection result (Unit IV, Example 7).

## Example 8 — Hinged-base portal frame (same loading)

- Column stiffness (base hinged) $= 3EI/4 = 0.75\,EI$; beam (symmetric) $= 0.333\,EI$. **$DF_{BA} = 0.75/1.083 = 0.692$; $DF_{BC} = 0.308$.**
- Balance at B: + 60 → BA = + 41.54, BC = + 18.46 → **$M_{BC} = -60 + 18.46 = -41.54$; $M_{BA} = +41.54$; $M_{AB} = 0$** (no carry-over to a hinge).
- Beam centre $BM = 90 - 41.54 =$ **+ 48.46 kN·m** ; $H = 41.54/4 =$ **10.38 kN**.

## Example 9 — Fixed-base frame, central point load P = 48 kN on the beam

- Same DF (0.75, 0.25). $FEM_{BC} = -PL/8 = -36$. Balance: + 36 → BA = + 27, BC = + 9.
- **$M_{BA} = 27$, $M_{AB} = 13.5$, $M_{BC} = -27$ kN·m** ; beam centre $= PL/4 - 27 =$ **+ 45 kN·m** ; $H =$ 10.13 kN.

## Example 10 — Beam stiffer than columns ($I_{beam} = 2\,I_{col}$)

- $K_{col} = 4EI/4 = EI$; $K_{beam} = 2\,(2EI)/6 = 0.667\,EI$ → **$DF_{BA} = 0.6$; $DF_{BC} = 0.4$** .
- Balance of + 60: BA = + 36, BC = + 24 → **$M_{BA} = 36$, $M_{AB} = 18$, $M_{BC} = -36$ kN·m** (compare

Unit IV, Example 8).

### Note for frames with sway

- If the frame sways (unsymmetrical loading, lateral loads), first analyse the **non-sway** frame (with an imaginary support), then give the sway displacement and correct the moments using the **shear equation**; this is beyond this unit's requirement.

## 8A. More Solved Examples — Beams

### Example 11 — Both ends fixed, two spans

**Problem**

ABC; A and C fixed; AB = 6 m (UDL 20 kN/m); BC = 4 m (central load 40 kN); EI constant.

- $K_{BA} = 4EI/6 = 0.667\,EI$; $K_{BC} = 4EI/4 = EI$ → **$DF_{BA} = 0.4$, $DF_{BC} = 0.6$**.
- FEM: AB = − 60, BA = + 60, BC = − 20, CB = + 20.

| Member end | AB | BA | BC | CB |
|---|---|---|---|---|
| DF | — | 0.4 | 0.6 | — |
| FEM | − 60 | + 60 | − 20 | + 20 |
| Balance B (unbalanced + 40) | | − 16 | − 24 | |
| Carry-over | − 8 | | | − 12 |
| Final | **− 68** | **+ 44** | **− 44** | **+ 8** |

- Reactions: span AB: $R_A = 60 + (68 - 44)/6 =$ **64 kN**, $R_B$(left) = 56 kN; span BC: $R_B$(right) = 20 + (44 − 8)/4 = 29 kN, $R_C = 20 - 9 =$ **11 kN**. **$R_B = 85$ kN**; check 64 + 85 + 11 = 160 √.
- √ matches slope-deflection: $EI\theta_B = -24$.

### Example 12 — Different EI in the two spans

**Problem**

Same loading, but BC has moment of inertia 2I (AB has I); A and C fixed.

- $K_{BA} = 4EI/6 = 0.667\,EI$; $K_{BC} = 4(2EI)/4 = 2\,EI$ → **$DF_{BA} = 0.25$, $DF_{BC} = 0.75$** (stiffer span attracts more moment).
- Unbalanced + 40 → balance: BA = − 10, BC = − 30; carry-over: AB = − 5, CB = − 15.

**Final moments**

**$M_{AB} = -65$, $M_{BA} = +50$, $M_{BC} = -50$, $M_{CB} = +5$ kN·m**

### Example 13 — Propped cantilever with a central point load

- A fixed, B roller: the far end is hinged so use modified FEM at A: $F'_{AB} = -PL/8 - \tfrac{1}{2}(PL/8) =$ **$-3PL/16$**. With one joint and a fixed end, $M_A = -3PL/16$ (hogging) directly; $R_B = 5P/16$ √.

### Reactions of Example 3 (three-span beam)

| Span | Left reaction | Right reaction |
|---|---|---|
| AB ($M_A$ = 62.31, $M_B$ = 55.38) | 60 + (62.31 − 55.38)/6 = 61.16 | 58.84 |
| BC ($M_B$ = 55.38, $M_C$ = 76.15) | 60 + (55.38 − 76.15)/6 = 56.54 | 63.46 |
| CD ($M_C$ = 76.15, $M_D$ = 0) | 60 + 76.15/6 = 72.69 | 47.31 |

- **$R_A$ = 61.16, $R_B$ = 115.38, $R_C$ = 136.15, $R_D$ = 47.31 kN** (sum = 360 = 3 × 120 ✓).

## 8B. Distribution Factors — Illustration

- Joint B of a frame: column BA ($I$, $h$ = 4 m, base fixed), beam BC on the left ($L$ = 6 m), beam BD on the right ($L$ = 5 m), all the same $E$ and $I$; far ends fixed.

| Member | Stiffness $4EI/L$ | Relative $K$ | DF |
|---|---|---|---|
| BA (column) | $4EI/4$ | 1.000 | 0.405 |
| BC (beam 6 m) | $4EI/6$ | 0.667 | 0.270 |
| BD (beam 5 m) | $4EI/5$ | 0.800 | 0.324 |
| Total | | 2.467 | 1.000 |

- If the far end of BD were hinged: $K_{BD} = 3EI/5 = 0.6$ (DF changes to 0.441, 0.294, 0.265 → sum = 1).
- **Stiffer members take the larger share** of the unbalanced moment.

## 8C. Kani's Method (brief)

- Kani (1947) — a **rotation-contribution method**; self-correcting iterative method for frames and beams.
- Rotation contribution of joint $i$ on member $ij$: $M'_{ij} = \mu_{ij} (\Sigma M^F_i + \Sigma M'_{ni})$, with rotation factor $\mu_{ij} = -\tfrac{1}{2} K_{ij} / \Sigma K_i$; final end moment $M_{ij} = M^F_{ij} + 2M'_{ij} + M'_{ji}$.
- Same answers as moment distribution; used mostly when many storeys are involved.

## 8D. Full Iteration for the Portal Frame (without using symmetry)

- The same frame as Example 7 ($h$ = 4 m, $L$ = 6 m, fixed bases, UDL 20 kN/m) solved by the plain method — columns $K = 4EI/4 = EI$, beam $K = 4EI/6 = 0.667\,EI$ → **$DF_{BA}$ = 0.6, $DF_{BC}$ = 0.4; $DF_{CB}$ = 0.4, $DF_{CD}$ = 0.6**.

| Step | AB | BA | BC | CB | CD | DC |
|---|---|---|---|---|---|---|
| DF | — | 0.6 | 0.4 | 0.4 | 0.6 | — |

| Step | AB | BA | BC | CB | CD | DC |
|---|---|---|---|---|---|---|
| FEM | 0.00 | 0.00 | -60.00 | +60.00 | 0.00 | 0.00 |
| Bal B(1) | | +36.00 | +24.00 | | | |
| CO(1b) | +18.00 | | | +12.00 | | |
| Bal C(1) | | | | -28.80 | -43.20 | |
| CO(1c) | | | -14.40 | | | -21.60 |
| Bal B(2) | | +8.64 | +5.76 | | | |
| CO(2b) | +4.32 | | | +2.88 | | |
| Bal C(2) | | | | -1.15 | -1.73 | |
| CO(2c) | | | -0.58 | | | -0.86 |
| Bal B(3) | | +0.35 | +0.23 | | | |
| CO(3b) | +0.17 | | | +0.12 | | |
| Bal C(3) | | | | -0.05 | -0.07 | |
| CO(3c) | | | -0.02 | | | -0.03 |
| Final | +22.50 | +45.00 | -45.00 | +45.00 | -45.00 | -22.50 |

- The cycles converge to **22.50, 45.00, – 45.00, + 45.00, – 45.00, – 22.50** — identical to the symmetric shortcut (Example 7) and to the slope-deflection result.
- Lesson: using **symmetry** replaces about 6 balancing steps by a single one.

### Check of the final moments

- Joint B: 45 – 45 = 0 √. Joint C: 45 – 45 = 0 √.
- Horizontal reaction: $H = (22.5 + 45)/4 = 16.88$ kN (equal and opposite at A and D √).
- Vertical reaction: $V = 60$ kN each ($\Sigma V = 120 = 20 \times 6$ √).
- Beam BC: centre BM = 90 – 45 = **+ 45 kN·m** ; points of contraflexure from $M(x) = 60x – 10x^2 – 45 = 0 \rightarrow x = 3 \pm 2.12 \rightarrow$ **0.88 m and 5.12 m** from B.

## 9. Reactions, Shear Force and Bending Moment Diagrams

- For every span: $R_{left} = R_{free} + (M_{leftend} – M_{rightend})/L$ (with hogging end moments positive).
- Draw SFD from the reactions; BM at any section = free BM – (linear variation of end moments).
- **Point of contraflexure:** where BM changes sign; **maximum sagging** where SF = 0.
- In frames, draw BMD on the **tension side** ; verify joint equilibrium and horizontal-force equilibrium at the bases.

## 10. Accuracy and Termination

- Continue cycles until the **carry-over moments are less than about 1%** of the FEM; the error in final moments is usually within 1–2%.
- Convergence is fast when the structure is not highly redundant; each carry-over halves the quantity.
- **Check:** (i) $\Sigma DF = 1$ at each joint; (ii) $\Sigma M$ at each joint = 0; (iii) $\Sigma V$ = total load.

## 11. Comparison — Slope-Deflection vs Moment Distribution

| Basis | Slope-deflection | Moment distribution |
|---|---|---|
| Nature | exact; equations | approximate (iterative) but converges to exact |
| Unknowns | joint rotations $\theta$ (and sway) | none solved directly |
| Equations | simultaneous equations | no equations |
| Effort | heavy for many joints | easier for many joints |
| Settlement | included through $\psi$ | included through extra FEM |
| Checks | substitute back | table totals, joint equilibrium |
| Developed by | G. A. Maney (1915) | Hardy Cross (1930) |

## 12. Advantages, Limitations and Related Methods

- **Advantages:** simple tabular work; no simultaneous equations; errors self-correct; physical meaning is clear; good for hand design of continuous beams.
- **Limitations:** convergence slows for large frames; sway frames need extra steps; not convenient for computer programming compared to matrix methods.
- **Kani's method:** a similar rotation-contribution iteration that is self-correcting.
- **Matrix stiffness method:** the computer-based extension ($K$ = member stiffness $4EI/L$, $3EI/L$, $6EI/L^2$, $12EI/L^3$).

## 13. Common Mistakes and Exam Tips

| No. | Point |
|---|---|
| 1 | $\Sigma DF$ at every joint = 1; a fixed support has $DF = 0$. |
| 2 | Use $3EI/L$ (and modified FEM) for a hinged far end and do **not** carry over. |
| 3 | Balancing moment = − (unbalanced moment) × $DF$ — sign change! |
| 4 | Carry over = ½ × balancing moment, to the **far end of the same member**. |
| 5 | Add settlement FEM = $6EI\Delta/L^2$ with the correct sign (B lower → anticlockwise). |
| 6 | Do not balance a fixed end; do not forget the known overhang moment. |

| No. | Point |
|---|---|
| 7 | Draw a neat table with joints, DFs, FEMs, cycles and a final sum row. |
| 8 | Verify by reactions: $\Sigma V$ = total load. |

## 14. Practice Problems with Answers

| No. | Problem | Answer |
|---|---|---|
| 1 | Two equal spans 5 m, UDL 24 kN/m, ends hinged | $M_B$ = 75 kN·m |
| 2 | Two equal spans 6 m, central loads 60 kN each, ends hinged | $M_B = 3PL/16$ = 67.5 kN·m |
| 3 | Fixed beam 6 m, UDL 20 kN/m, MDM check | $M_A = M_B$ = 60 kN·m |
| 4 | Portal frame h = 4 m, L = 6 m, fixed bases, UDL 30 kN/m on the beam | $M_{BA}$ = 67.5; $M_{AB}$ = 33.75 kN·m |
| 5 | Two equal spans 6 m, w = 20, B settles 10 mm, EI = 30000 | $M_B$ = 65 kN·m |
| 6 | Three equal spans 6 m, UDL 10 kN/m, ends hinged | $M_B = M_C = wL^2/10$ = 36 kN·m |

## 15. Quick Revision — Unit V

**Key formulae**

$K = 4EI/L$ (far end fixed), $3EI/L$ (hinged) | $COF = ½$ | $DF = K/\Sigma K$, $\Sigma DF = 1$ | Balancing moment = − unbalanced × DF | Carry-over = ½ × balancing moment | Modified FEM (hinged far end): $F - ½F_{far}$ | Settlement FEM = $\pm 6EI\Delta/L^2$ | Symmetric span: $K = 2EI/L$.

### Possible exam questions

1. Explain the moment distribution method; define stiffness, carry-over factor and distribution factor.
2. Analyse a continuous beam by the moment distribution method and draw SFD and BMD.
3. Analyse a continuous beam with settlement of supports by MDM.
4. Analyse a single-bay portal frame without sway by MDM; draw the BMD.
5. Compare slope-deflection and moment distribution methods.